Supplementary Material: SHARE-Eval: A Stage-Level Benchmarking Framework for Multi-Agent Debate and Consensus in Data Workflows

S1. Overview

We use *Du-MAD* as a label for the multiagent debate protocol of Du et al. [1].

Generation supplies initial candidates; final-answer generation uses a candidate pool to produce the workflow result within Final Decision. In fixed-pool comparisons, generation abbreviates final-answer generation. Pair F1 abbreviates candidate-pair F1, which scores judgments on supplied record pairs; cluster F1 scores relations implied by record groups.

Sections S2–S4 provide the replay proof and contrast classifications, extractor audits, implementation records, populations, protocol deviations, and decoding controls. They support the methodology, suite, and setup in Sections III–IV.

The remaining sections follow the four result chapters: peer information, content, and source (Section S5); candidate losses, output rules, and grouping (Section S6); stage replacements (Section S7); and additional computation (Section S8). These include complete configurations, failure counts, execution costs, and test-suite checks. Each section identifies the main-text result it supports. The reproducibility package provides the protocol and pipeline comparisons at https://github.com/wtwinlab/AgentEAB.

S2. Methodology: Replay Validity and the Extractor

Replay reproduces deployment when deterministic output transformations can be recomputed and no retained model call or continuation decision reads a changed input (Lemma 1), and in two blind audits a human accepted 49 of 50 recovered queries, and 50 of 50 that decide a Du-MAD vote, as the intended final answer.

A. Proof of Lemma 1

Setup. A run of unit u is a sequence of stages $k = 1, \ldots, K$; stage k produces $x_k = f_k(\rho_k(h_{k-1}), \omega_k)$, where h_{k-1} collects previous outputs, control decisions, and state, ρ_k is the stage's read set, and ω_k is exogenous randomness. The outcome is $Y = g(h_K)$. Replay R_u computes $x'_j = o'(\rho_j(h_{j-1}))$ on the recorded prefix, recomputes downstream deterministic output transformations, and retains later recorded model outputs and continuation decisions. Recomputed transformations include the return rule, grouping, and scoring. The deployed counterfactual O_u runs the workflow with o' at stage j. Walking forward from j, propagate the replacement through the recomputed transformations. Set $V_u = 1$ when a retained model call or continuation decision receives an input different from its recorded input. Up to the first such read, replay and deployment agree, although deterministically propagated values may differ from the original record. The first divergent read is therefore identified from recorded inputs and deterministic propagation. A divergence is *recoverable* when its first differing read is a continuation decision under which the replacement terminates the run and its outcome can be computed without an unrecorded continuation. Write V_{rec} and V_{unrec} for recoverable and unrecoverable divergences, so that $\{V = 1\} = V_{\mathrm{rec}} \cup V_{\mathrm{unrec}}$.

(i) Exact replay. At stage j, replay and deployment apply the same deterministic replacement to the same recorded prefix, so their updated outputs agree. Proceed by induction over $k > j$. A recomputed deterministic transformation receives the same updated history in replay and deployment and therefore produces the same output in both; that output need not equal the original recorded output. At a retained model call or continuation decision, $V_u = 0$ means that its input agrees with its recorded input. Under A1, the unchanged stage function produces the recorded output retained by replay. The two updated histories therefore agree throughout, giving $R_u = O_u$. A2 makes the replacement computable from recorded inputs, A3 excludes hidden dependencies, and A4 treats a joint operation such as grouping as one unit.

(ii) Bound. R and O differ only on $\{V = 1\}$, so for outcomes in $[0, 1]$, $|E[R - O]| \le E|R - O| \le P(V{=}1)$ (the coupling inequality on the pair built from the same ω).

(iii) Interval. For $Y \in [0, 1]$, $E[O]$ lies in $[E[R\,1\{V{=}0\}] + E[\tilde{Y}\,1\{V_{\mathrm{rec}}\}],\ E[R\,1\{V{=}0\}] + E[\tilde{Y}\,1\{V_{\mathrm{rec}}\}] + P(V_{\mathrm{unrec}})]$, where $\tilde{Y}$ is the exactly computed outcome on recoverable divergences; this is Manski's worst-case bound applied to the unrecoverable units.

Without A1. Take ω' with the same law as ω and unchanged stage functions. Then $O(\omega')$ and $O(\omega)$ have the same distribution, so $E[O]$ is unchanged and (ii)–(iii) still bound $|E[R] - E[O]|$. Here V is defined under the common-randomness coupling, rather than by disagreement between two independent executions. If $V_u = 0$ for every realisation, the induction above gives $R(\omega) = O(\omega)$ under that coupling. Since $O(\omega')$ has the same distribution as $O(\omega)$, replay and fresh deployment are equal in distribution.

B. Classification of the 24 Replacement Contrasts

Table S1 classifies the 24 protocol and pipeline replacement contrasts under Lemma 1: an *exact* contrast reproduces the online result exactly after deterministic output transformations are recomputed, with no changed input to a retained model call or continuation decision; a contrast that holds *on average* compares two deployed conditions or a single model draw; a *bounded* contrast includes BCCS, whose stop test reads its own extracted answers, and carries the share of runs that diverge and the worst-case interval. Shares of diverging runs are given per condition where they differ; bounds use 198 BIRD questions with two repeats averaged per question, and the 95% bootstrap intervals widen bounds by 1–5 pp.

C. An Online Check of the Bound

This comparison extends the BCCS/Qwen2.5-7B check in Section IV. Replay and online executions have independent generated responses and different serving configurations, so they do not share the exogenous randomness required for path-wise equality. Run-level outcome disagreements are 11.4% with peers and 9.8% without; repeat-to-repeat disagreements within a collection range from 7.6% to 13.6%. Thus run-level mismatches also contain ordinary decoding variation. The bounds allow 79.0 and 53.0 pp, far exceeding the observed 0.25-pp accuracy gaps; this check concerns one configuration and does not establish tightness.

Running BCCS online on Qwen2.5-7B with execution voting inside the protocol, so that the vote is deployed rather than replayed, gives outcomes inside the lemma's interval in both conditions: the online run differs from its replay by -0.25 pp with peers (95% CI -3.03 to $+2.53$) and by $+0.25$ pp without peers (-3.28 to $+3.79$); among the 65 no-peer tasks whose only divergence is recoverable, the online run is 3.08 pp below its replay (-6.15 to -0.77, over questions). The peer interaction effect of this online run is reported with the scale study (Section S8-C).

D. Audit of the Lenient Extractor

When the protocol's extractor finds no answer, the lenient extractor takes the last SQL code block; if none exists, it takes the last standalone query. In 17,350 BIRD calls only the lenient extractor finds SQL. A blind human audit of 50 of these calls, sampled without regard to the outcome, accepted 49 recovered queries as the intended final answer (95% CI 89.4–99.9%). Two agent judges, GPT-6 Astra in ChatGPT Web and Claude Opus 5.5 in Claude Web, agreed with the annotator on 50/50 and 48/50 of the same items; on 450 further items they accepted 438 (97.3%, 95% CI 95.4–98.6%). A second blind audit by the same annotator took 50 recovered queries that decide an execution vote in the Du-MAD runs on the 1,531 questions, stratified by backbone and condition, and accepted all 50 (Section S6-F). The audits are sample checks of what the lenient extractor recovers, not a re-scoring of every call; Section S6-F bounds what extractor errors could change in the BIRD debate effects.

E. Effects, Transitions, and Merges

The definitions below underlie the effects, transitions, and merge condition of Section III-C. *Paired effect.* For outcome Z, n paired tasks, and condition $k \in \{0, 1\}$ (baseline, alternative), $\overline{Z}_i^{(k)}$ averages outcomes within task i: repeated observations for a task or question, or question means for a database. The estimated mean effect is

$$\widehat{\Delta}_Z = \frac{1}{n} \sum_{i=1}^{n} \big(\overline{Z}_i^{(1)} - \overline{Z}_i^{(0)}\big). \tag{1}$$

Paired outcome transitions. Let W, C partition scored baseline outcomes into failures and successes, and S contain alternative successes. Correction is $c = |W \cap S|/|W|$ and regression $r = |C \setminus S|/|C|$. On this common binary population with baseline accuracy q,

$$\Delta Q = (1 - q)c - qr, \tag{2}$$

so recovered failures minus lost successes equals the accuracy effect in Eq. (1). These are paired-condition outcomes, not successive revisions of one answer.

When a merge raises cluster F1. Merging two groups adds every cross-group pair. If $b > 0$ new pairs include τ true matches and the original cluster F1 is $F = 2T/(P + G)$, the score becomes $2(T + \tau)/(P + b + G)$, which exceeds F exactly when $\tau/b > F/2$: the true-match fraction of the added relations exceeds half the original score, including when existing groups contain errors.

Added and removed relations. Let E_0, E_1 be the implied pairs before and after a change and E^* the reference pairs. With $D^+ = E_1 \setminus E_0$ and $D^- = E_0 \setminus E_1$, the true and false counts are $a_T = |D^+ \cap E^*|$, $a_F = |D^+ \setminus E^*|$, $r_T = |D^- \cap E^*|$, $r_F = |D^- \setminus E^*|$, and with P original implied pairs, T of them true, and G reference pairs, the score after the change is $F_1' = 2(T + a_T - r_T)/(P + G + a_T + a_F - r_T - r_F)$.

TABLE S1: Replay conditions and deployment interpretation of the 24 replacement contrasts.

ID	Contrast	Class	Share of diverging runs	Replay estimate (pp)	Deployment bound (pp)	Interpretation
R1	final answer − lenient execution vote over the initial answers (6 interactive configurations)	exact	0 (no later step)	-10.86 [-13.13, -8.71]	point	exact replay
R2	matching final answer − vote over the initial answers, lenient extractor on both sides (16 with-peer configurations)	exact	0	13 of 16 within 2.6 pp	point	exact replay
R3	MoA aggregator − vote over its references (exact-match or execution vote; ties to the first agent, the last agent, or at random)	exact	0	final − execution vote -0.51 / -14.65 / -6.31 (first / last / random)	point	exact for the stated tie rule
R4	extractor, vote, and both at the last round; Du-MAD and MAD-M^2 (4 configurations)	exact	0	extractor $+5.56$ [$+4.17$, $+7.01$]; vote $+2.97$ [$+1.96$, $+4.04$]; both $+8.78$ [$+6.82$, $+10.80$]	point	exact replay
R5	extractor, vote, and both at the last round; BCCS (2 configurations)	bounded	with peers: Llama-3-8B 0.854 (unrecoverable 0.745; tasks with any divergence, 95% CI 0.94–0.99); Qwen2.5-7B 0.790 (0.523; 0.88–0.96)	both: Llama $+8.59$ [$+5.30$, $+11.87$]; Qwen $+4.29$ [$+1.77$, $+6.82$]	Llama [-74.75, $+81.57$] (CI [-79.80, $+85.35$]); Qwen [-43.18, $+59.34$] (CI [-49.24, $+64.65$])	evaluated on recorded responses; deployment sign not identified
R6	mean of six configurations: extractor $+4.17$, vote $+2.86$, both $+8.00$	bounded	BCCS contributes 2 of the 6 configurations	extractor $+4.17$ [$+3.03$, $+5.35$]; vote $+2.86$ [$+2.02$, $+3.75$]	extractor [-17.51, $+25.38$] (CI [-19.36, $+26.94$]); vote [-20.83, $+27.10$] (CI [-22.31, $+28.41$]); both [-13.80, $+29.34$] (CI [-16.08, $+31.10$])	deployment claim stated for Du-MAD and MAD-M^2 (the four-configuration row above); the six-configuration mean evaluated on recorded responses
R7	round switch, joint extractor+vote+round, non-additivity ($+8.54$ vs $+3.70$)	bounded (mean of 6); exact (mean of 4)	BCCS with peers: Llama 0.818, Qwen 0.503	non-additivity, mean of 6: $+4.84$ [$+3.28$, $+6.44$]; mean of 4: $+4.10$ [$+2.27$, $+5.93$], over databases [1.89, 5.81]	mean of 6 [-40.61, $+50.21$] (CI [-43.43, $+53.16$])	exact for the four-configuration mean ($+4.10$); the six-configuration value evaluated on recorded responses
R8	Δ_{peer} (lenient extractor, execution vote), mean of 6 $= -1.52$	bounded (mean of 6); on average (mean of 4)	with peers / no peers: Llama 0.854 / 0.616; Qwen 0.790 / 0.530	mean of 6: -1.52 [-2.99, -0.04]; mean of 4: -1.96 [-3.85, 0.00], over databases [-3.35, -0.38]	mean of 6 [-16.29, $+17.59$] (CI [-18.27, $+19.70$]): sign not identified	Du-MAD and MAD-M^2: on 198 questions the question interval's upper end is 0.00 and the database interval's -0.38 pp; on 1,531 questions the peer interaction effect is detected, and the Du-MAD loss is detected on both the last-round difference and the peer interaction effect (Table S17); BCCS evaluated on recorded responses
R9	later rounds, last round − initial answers, with peers (lenient extractor, execution vote) $= -0.55$	bounded (mean of 6); exact (mean of 4)	as the peer-interaction row above, with peers	mean of 6: -0.55 [-1.77, $+0.63$]; mean of 4: -0.38 [-1.96, $+1.20$]	mean of 6 [-4.71, $+16.41$] (CI [-6.14, $+18.31$])	scope as the peer-interaction row
R10	Δ_{peer} (protocol's extractor, exact-match vote) $+3.07$ and its tie variants; MAD-M^2/Qwen2.5-7B $+8.33$ when ties go to the first agent	exact (the MAD-M^2 tie dependence); bounded (mean of 6)	ties to the last agent, with peers: Llama 0.828, Qwen 0.538	mean of 6, ties to the first agent $+3.07$ [$+1.22$, $+4.92$]; mean of 4 $+3.03$ [$+0.95$, $+5.11$]	mean of 6 [-14.65, $+22.56$] (CI [-16.92, $+24.58$])	tie-order dependence is exactly replayable
R11	BCCS baseline configuration (missing labels dropped) − final answer: $+6.6$ to $+7.3$	bounded	with peers: Llama 0.818 (unrecoverable 0.818; tasks with any divergence 0.90–0.97); Qwen 0.503 (0.503; 0.63–0.76)	Llama $+6.57$ [$+4.29$, $+9.09$]; Qwen $+7.32$ [$+4.55$, $+10.35$]	Llama [-5.05, $+76.77$] (CI [-7.32, $+81.06$])	evaluated on recorded responses; mechanism: runs stop on a shared missing label
R12	MAC-SQL Refiner − the Decomposer's query (4 configurations)	exact	0	$+1.83$ / $+3.46$ BIRD; $+0.39$ / $+0.97$ Spider	point	exact replay

ID	Contrast	Class	Share of diverging runs	Replay estimate (pp)	Deployment bound (pp)	Interpretation
R13	CHESS Unit Tester vs self-consistency, Valid, Nonempty	exact	0	Unit Tester − self-consistency -1.36 $[-5.71, +2.88]$	point	exact replay
R14	ComEM selector − the ranker's own decision (2 selectors)	exact (for the logged selector draw; the ranker is deterministic)	0	$+5.19$ / $+1.31$	point	temperature 0, fixed seed
R15	SQL rules on fixed pools (Unfiltered, Valid, Nonempty; Spider Nonempty − Valid)	exact	0	Section VII-B	point	exact replay
R16	generation or LLM selection vs rules on the fixed pool	on average (single draw)	n/a	Section VIII-A	n/a	estimates from one model response
R17	grouping rules on fixed judgments (center vs components; development-selected vs fixed; 15 rules)	exact (block level)	0	Section VII-C	point	exact replay
R18	sharing decisions → 23 changed judgments → cluster F1 -3.05 pp (Settlements)	on average (two deployed conditions); an edge-level bound does not apply	changed-edge share 0.49% vs $\lvert\Delta$F1$\rvert$ 3.05 pp	-3.05 pp	relation-level interval $[-3.305, 6.61]$ pp	evaluated over group relations
R19	QA readable-only plurality vs the original return rule ($+3.15$ pp, 1,680 sets)	exact	0	$+3.15$ [1.90, 4.58]	point	exact replay
R20	matching return rule → all-round majority (Du-MAD, MAD-M^2, BCCS); FORD	exact	0	moves up to 1.98 pp (0.004–1.98 across the three tie rules); MAD-M^2/Settlements $+2.21 \rightarrow +0.23$ / $+0.45$ / $+0.78$ (interval then covers 0); Du-MAD/Settlements $-0.14 \rightarrow +0.10$ (ties to the last agent / kept)	point	exact replay; the effect moves up to 1.98 pp
R21	ReConcile confidence re-parse (a QA protocol outside the body's matrix)	exact	0	pair F1: 0.133 → 0.871 (WDC; 0–1 scale)	point	additional extractor example
R22	deployed fixed-input contrasts: QA fixed peers, stronger peers, execution feedback ($+3.66$)	on average	n/a	Sections V-B and VIII	n/a	compares deployed outcomes
R23	end-to-end with peers − no peers (Fig. 3) and the no-peer factor	on average	n/a	-4.5 to $+28.5$	n/a	compares deployed outcomes
R24	scale table Δ_{peer} at Qwen2.5-14B, 32B, 72B (Table X)	bounded (BCCS); on average (Du-MAD, MAD-M^2)	BCCS 14B: with peers 0.684 (unrecoverable 0.384); no peers 0.354 (0.015); 32B: with peers 0.803, no peers 0.535; 72B shares not computed	BCCS 14B -4.04 $[-8.59, +0.51]$; 32B -0.25 $[-3.54, +3.03]$; 72B $+1.01$ $[-1.52, +4.04]$	14B $[-17.17, +22.73]$ (CI $[-22.98, +28.79]$); 32B $[-11.11, +21.46]$ (CI $[-15.40, +27.02]$); 72B not computed	BCCS scale comparisons use recorded responses

S3. Implementation Records and Reproduction

The implementation in Section IV connects collection, offline comparisons, and recomputation of reported results. Its inputs differ by reproduction level: result objects consume scored tables; record-level analysis consumes the underlying responses and control records; new collection additionally requires the tested models and benchmark inputs.

A. Information Retained by the Collectors

Table S2 summarizes the information used by the analyses. The adapter and experiment determine which fields are available. Protocol-specific state supplements the common identifiers when it influences a later decision.

TABLE S2
Execution information used by the suite.

Information	Purpose and availability
Task and configuration	Execution records and study manifests jointly identify tasks, protocols, models, conditions, and repeats, with available configuration and source fingerprints.
Model responses	Instrumented adapters retain sent messages, returned answer text, extracted candidates, and agent/round identifiers. Returned text need not include a separately exposed reasoning channel.
Continuation state	BCCS records peer sets, beliefs, clusters, and stopping outcomes. Another protocol requires the state read by its own continuation rules.
Usage and timing	Per-call usage is available for instrumented adapters; protocol wrappers can provide endpoint-counter differences. The analysis states the granularity and treatment of missing usage.
Task outcomes	SQL result signatures support voting; separate reference-based verdicts score correctness. Stage analysis retains pipeline-specific comparison semantics.

B. Reproduction Levels

The shared result-object entry point recomputes 29 objects and checks 3,143 printed numerical values against their computed outputs, together with 101 qualitative statements in the manifest. Each object has scored inputs, a computation, expected values, and a manifest linking quantities to output keys. The check compares these declarations with computed values at the stated precision. With corresponding manuscript sources, the check can additionally search for the declared printed forms.

A second entry point reruns three checks supported by the included inputs: database-cluster bootstrap output, deployment bounds, and the grouping-bound calculation. It compares their outputs with stored digests. These checks exercise the supplied scored-input paths. Response replay uses execution records and cached SQL outcomes, or the benchmark databases when outcomes must be recomputed. Reconstructing BCCS divergences also requires its control records. SQL pipeline re-execution additionally requires the benchmark databases. New protocol executions use the corresponding collection drivers and model endpoints.

The reproduction instructions identify the relevant entry points and additional data for each study. Protocol-specific scripts and shared helpers provide the current collection and analysis paths; extending the suite requires the corresponding input, scoring, dependency, and output declarations.

S4. Setup: Populations, Deviations, and Decoding Checks

The coverage summarized in Section IV is expanded in Table S3, the protocol deviations in Table S4, and the decoding settings below; none of the body's QA sharing conclusions changes when only the first response generated for each input is used. The no-peer condition removes each peer block in full, together with the clauses that refer to peers, so no fragment of a peer reply remains in the prompt. The BIRD debate questions are 198 of 200 drawn by round-robin over database and difficulty strata from the BIRD development set (68, 68, and 62 simple, moderate, and challenging), with the two whose reference query exceeds 5 s excluded; the four Du-MAD and MAD-M^2 configurations also run on the remaining 1,333 development questions (one further question with a timing-out reference query excluded).

TABLE S3
Experiment matrix.

Study	Task, data	Models	Units	Conditions	Section
Protocol matrix	BIRD; WDC, Settlements; GSM8K, StrategyQA, MMLU-Pro, GPQA	five protocols; every agent Llama-3-8B or Qwen2.5-7B; FORD and MoA mixed-model rosters	198 questions; 1,000 pairs per dataset; 100 tasks per benchmark	with peers, no peers, answers only	V, VI
BIRD debates, controls	BIRD	Du-MAD, MAD-M^2; Llama-3-8B, Qwen2.5-7B agents	198 questions; the remaining 1,333	placebo peers; temperature 0.7; unified answer format; with, without peers	VI
BIRD debates, scale	BIRD	Du-MAD, MAD-M^2, BCCS; Qwen2.5-14B, 32B, 72B agents	198 questions	with, without peers	VIII
BIRD debates, families	BIRD	Du-MAD, MAD-M^2; 16 models	198 questions	with, without peers	VIII
QA with fixed peers	GSM8K, StrategyQA, MMLU-Pro, GPQA	Qwen3-32B, gpt-oss-120b, Gemma-4-26B-A4B, Qwen2.5-7B recipients; Llama-3-8B, Qwen2.5-7B, Qwen3-32B peers	480 tasks, three positions	no peers, answers only, full responses	V
Shared content	DBLP–ACM, Amazon–Google	the three recipients; Qwen2.5-7B peers	800 and 1,000 test pairs	four shared inputs	V
SQL candidate pools	BIRD, Spider	Qwen2.5-7B, 32B, 72B; the three recipients; DeepSeek-V4-Pro	1,534 and 1,034 questions	vote rules; selection; generation	VII, VIII
Grouping	Settlements, WDC, MusicBrainz	Qwen2.5-7B and 32B judgments	4,719, 2,615, 32,199 pairs	15 grouping rules	VI, VII
Pipelines	MAC-SQL: BIRD, Spider; CHESS: BIRD; ComEM: eight datasets	Qwen3-32B, Qwen2.5-72B; Qwen3-32B; Qwen3-32B, Qwen2.5-7B	1,534 and 1,034; 147; 3,200 query records	stage against rule	VII
QA debate traces	GSM8K, StrategyQA, MMLU-Pro, GPQA, SQuAD2	eight protocol variants; Llama-3-8B, Qwen2.5-7B, Mistral-7B	10,295 traces	as logged	VI

A. Replicates and Statistical Families

The weighting and detection rules are defined in Section IV; the following specifies the replicate counts and families behind those comparisons.

Qwen2.5-7B/32B generation uses three generation repeats, retaining 1,533/1,532 BIRD questions, respectively, and 1,034 Spider questions with all three available. Primary cross-family SQL generation comparisons use the first response generated for each input on the full benchmarks. Qwen3's matched-input selection and generation use two matched generation repeats per question in both pools, averaged within questions before equal database weighting; its execution-summary selector uses one. DeepSeek-V4-Pro selects with the generation prompt and generates once per question on the four-model pool. Table IX uses the first response generated for each input.

TABLE S4
CONFIGURATIONS AS RUN AND DEVIATIONS FROM THE PUBLISHED CODE.

Protocol	As run
All five	The no-peer condition removes the peer replies and the clauses that refer to them from every prompt and keeps rounds, stopping, and return rules, so Du-MAD revises its own previous answer and MAD-M^2, which rebuilds each prompt, solves the question again; the answers-only condition shares extracted answers and falls back to the response text when extraction fails; the extracted SQL answers are compared by exact text match; two repeats per configuration.
FORD	Greedy decoding (temperature 0); the matrix's mixed-model roster; by design, its first turn sees a peer.
MoA	Fixed responses from other agents; answers the extractor cannot classify are rendered as "unclear"; without peers the aggregator receives no other agent's response; Gemma-4-26B-A4B in the roster.
Du-MAD	Greedy decoding (the official MMLU code leaves temperature unspecified [14]), so only 4–8% of BIRD runs have initial queries with two or more distinct execution results; the evaluated SQL implementation accepts boxed answers only.
MAD-M^2	Objective filter, which shows each agent one peer reply chosen by the probability the model assigns to it, as published; for gpt-oss agents the filter reads the log-probabilities of the final channel.
BCCS	As published; its stop test, no-consensus selection, and choice of peers read the extracted answers.
MAC-SQL	Qwen3-32B and Qwen2.5-72B-AWQ; the rule that replaces the Refiner keeps the Decomposer's query.
CHESS	Commit `3d6e835f`; 20 generated unit tests plus one fixed test (the paper reports ten); Qwen3-32B without thinking for every engine; local bge-small embeddings replace the embedding service; recorded temperatures 0.5 for revision and 0.8 for test generation and evaluation; 40,960-token context, 4,096-token output, no per-request seeds; read-only database connections.
ComEM	Flan-T5-XL ranker; Qwen3-32B or Qwen2.5-7B selector at temperature 0 with a fixed seed; a query record counts as correct only when all ten of its match decisions are correct.

Holm corrections [31] apply within the stated families: nine cross-family QA comparisons; all 18 cross-family SQL contrasts jointly, with a recipient's three contrasts as a secondary family; six within-pool operation contrasts on the Qwen3/GLM pool and, separately, six cross-pool changes. Separate families cover peer-source, selector, CHESS, Qwen2.5-7B input, five Qwen2.5 scale contrasts, and cross-family debate models. The change in the answers-only versus full-response effect across peer sources for the Qwen2.5-7B QA recipient is descriptive.

The sampled factorial comparison and the full-development comparison use separate Du-MAD no-peer collections on the 198 shared questions. Their outputs differ on 53 of 792 runs. The corresponding four-configuration peer interaction effects are -1.96 and -1.83 pp; neither is detected. Within the 1,333 other development questions, paired with/no-peer executions give -1.30 pp over four protocol–backbone pairs, detected. No paired contrast pools the two no-peer collections. Answer-only and budget comparisons use a separate matched collection on the same 198 questions.

BCCS dependency checks reproduce 3,553 verdicts, 2,688 stopping decisions, and 1,969 peer choices on the responses underlying the sampled factorial comparisons. Of 24 protocol and pipeline contrasts, 12 satisfy exact-replay conditions, four hold on average, three require bounds, and five support deployment conclusions for Du-MAD and MAD-M^2 alone.

B. Decoding, Pools, and Selector Inputs

QA sharing. Each of the three primary recipients decodes 2,880 inputs per condition (two sets of peer responses, three positions, 480 tasks) with a 32,768-token output budget that includes reasoning; Qwen3 uses thinking mode and gpt-oss medium reasoning. Qwen3 and gpt-oss decode a fixed subset a second time (556 no-peer, 649 answers-only, and 575 full-response inputs); Gemma decodes once. Conditions are paired by task, peer set, position, and decoding repeat before differences are averaged within tasks. The Qwen2.5-7B recipient uses a 1,024-token output limit, so cross-recipient differences do not isolate recipient capability.

Candidate pools. The four-model pool holds one query each from Llama-3-8B and Qwen2.5-7B/14B/32B; the Qwen3/GLM pool holds two samples each from Qwen3-32B and GLM-4.7-Flash. Both use temperature 0.6, top-p 0.95, top-k 20, min-p 0, and a 12,288-token output limit including reasoning, with a distinct seed per question, source, and sample. SQL is extracted from the reply's answer, after any reasoning; a reply with no extractable query remains a failed candidate. Selection uses execution status and result sets, never reference labels.

Selector inputs. The selector given an execution summary receives the database identifier, the question, benchmark evidence when present, the candidate SQL, and a summary of each candidate's execution (status, row count, column names, up to three rows, and any error text, each truncated). Generation receives the schema and the full execution results, including elapsed time and up to three rows without truncation. The LLM selector with full inputs reuses the generator's prompt with three output instructions changed, which adds exactly 11 tokens; schema, hints, SQL, execution results, candidate order, decoding, budget, and seed match generation. Table S5 varies candidate order and reasoning for the Qwen3-32B selector given an execution summary: condition A lists Qwen2.5-7B, Llama-3-8B, Qwen2.5-14B, and Qwen2.5-32B in a fixed order; B applies a seeded per-question permutation; A and B disable thinking at temperature 0 with a 256-token limit; C uses B's permutation with thinking, temperature 0.6, top-p 0.95, top-k 20, min-p 0, and a 32,768-token limit, matching generation.

Invalid candidate choices (5 in A, 2 in B, none in C) count as failures. After randomization, B selects the first position 40.5% of the time on BIRD and 55.6% on Spider, and C 40.7% and 51.6%, which supports a position preference without excluding a source preference.

Condition C consumes 1,102,863 input and 3,638,952 output tokens (3,623,544 reasoning tokens) over 2,568 questions, with mean latency 40.3 s on BIRD and 35.5 s on Spider on two H200 GPUs serving 96 concurrent requests; A and B consume 1,113,135 input tokens each and 7,971 and 7,873 output tokens. Accuracy is 54.20, 53.73, and 55.25% for A, B, and C on BIRD and 76.23, 76.97, and 79.28% on Spider, with equal database weights; of the eight contrasts in Table S5, only generation minus C on BIRD passes the Holm correction.

TABLE S5
SELECTOR CONTROLS: CANDIDATE ORDER AND REASONING (PP).

Benchmark	Contrast	Effect	95% CI
BIRD	B − A	-0.46	$[-1.50, 0.53]$
BIRD	C − B	$+1.51$	$[0.20, 2.59]$
BIRD	C − Nonempty	$+0.98$	$[-0.47, 2.13]$
BIRD	Generation − C	$+2.64$	$[1.05, 4.05]$
Spider	B − A	$+0.74$	$[-0.84, 2.47]$
Spider	C − B	$+2.31$	$[0.23, 4.26]$
Spider	C − Nonempty	$+0.58$	$[-1.78, 2.58]$
Spider	Generation − C	-1.01	$[-2.92, 0.69]$

C. Using Only the First Response per Input

Using only the first response generated for each input in every QA contrast gives a common 2,880-observation population per recipient and changes no difference by more than 0.25 pp and no Holm decision (Table S6).

TABLE S6
QA SHARING EFFECTS USING ONLY THE FIRST RESPONSE PER INPUT (PP).

Recipient	Contrast	Difference [95% CI]
Gemma	Full − no peers	-4.17 $[-6.32, -2.01]$
Gemma	Answers − no peers	-2.85 $[-4.83, -0.97]$
Gemma	Answers − full	$+1.32$ $[-0.38, 3.02]$
gpt-oss	Full − no peers	$+0.17$ $[-2.08, 2.43]$
gpt-oss	Answers − no peers	$+6.18$ $[4.20, 8.26]$
gpt-oss	Answers − full	$+6.01$ $[4.24, 7.74]$
Qwen3	Full − no peers	-1.39 $[-4.03, 1.25]$
Qwen3	Answers − no peers	$+1.94$ $[-0.35, 4.20]$
Qwen3	Answers − full	$+3.33$ $[1.25, 5.42]$

Task weights; one nine-contrast Holm family.

S5. Q1: Effects of Peer Information

Sharing changes final results by -4.5 to $+28.5$ pp (Fig. 3), and FORD's largest gains concentrate in runs whose no-peer condition exhausts its rounds. Fixed-message comparisons distinguish the effects of shared content and peer source on individual recipients.

A. FORD's Stopping Rule and Exhausted Ties

Without peers, FORD's stopping rule never fires on 16.7% of WDC and 20.2% of Settlements runs, and on 14–96% across QA and BIRD (GSM8K 83.5%, MMLU-Pro 58%, GPQA 56%, BIRD 95.5%); with peers it still fails to fire on 74% of BIRD runs. An exhausted run ties only for binary answers: with consecutive agreement the last prediction is the strict all-round majority, whereas an exhausted even-length run ties. On WDC, returning the first tied prediction changes 333 no-peer runs and none of the 2,000 with-peer runs and moves the sharing effect from $+26.23$ to -0.90 pp; returning the last prediction or keeping the protocol's answer preserves $+26.23$ pp, and the protocol itself returns the last prediction. On the 1,667 paired runs in which the no-peer condition does not tie, sharing adds 0.50 pp of F1; the 333 tied runs account for 7.25 of the 7.35-pp accuracy gain (accuracy adds across the two groups, whereas F1 must be recomputed from the combined counts). The same runs also set FORD's resource gains: on WDC, mean calls fall from 6.00 to 2.23 in the tied runs and from 2.01 to 2.00 elsewhere. On GSM8K, sharing gains 28.50 pp with about 2.7 fewer calls and 389 fewer output tokens per task, and on MMLU-Pro 13.50 pp with fewer input and output tokens. For MAD-M^2 with Llama-3-8B, counting every matching reply the extractor cannot resolve as wrong or as right gives bounds of $[-7.09, +2.46]$ pp on WDC and $[-4.79, +0.72]$ pp on Settlements, which is why Fig. 3 marks these configurations.

B. QA Transitions under Answers-Only Sharing

Table S7 separates recovered errors from lost successes for the answers-only condition of Table III; its paired population averages matched positions and generation repeats within tasks, so the no-peer accuracy q differs slightly from the all-output value in the body. Gemma starts with fewer errors, corrects a smaller fraction of them (25.34%), and loses a larger fraction of its correct answers (8.96%) than gpt-oss (40.88% and 6.10%); Qwen3 has the highest regression rate (13.92%) but recovers enough errors for a positive estimate. With both peers wrong, the regression rates are 29.75%, 19.63%, and 40.77% for Gemma, gpt-oss, and Qwen3. The 2,880 distinct peer answers have 54.51% accuracy, peers are less accurate than the recipient in all 12 recipient–benchmark comparisons, and in 25–32% of the 960 task and peer-set combinations per recipient the recipient is correct while the peer majority is wrong.

TABLE S7
ANSWERS-ONLY SHARING: PAIRED NO-PEER ACCURACY AND TRANSITIONS.

Recipient	q	c	r	Recovered	Lost	ΔQ
Gemma	82.19	25.34	8.96	4.51	7.36	-2.85
gpt-oss	73.82	40.88	6.10	10.70	4.50	$+6.20$
Qwen3	70.16	40.08	13.92	11.96	9.77	$+2.19$

q, c, r in percent (Eq. (2)); recovered, lost, ΔQ in pp.

C. What Peers Share

The confusion counts and content levels below support Section V-C. *Field comparisons on Amazon–Google.* On the same 1,000 test pairs, Qwen3's errors concentrate in missed matches (Table S8). With a peer decision and field comparisons, 143 correct no-peer judgments become incorrect and 41 incorrect ones become correct; 125 of the losses are true matches changed to non-match and one is unreadable. The shown peer rejects 224 of the 500 true matches, and both the losses (140 of 143) and the corrections (41 of 41) mostly follow the shown decision, a post-hoc association because the peer's decision and its field evidence were not varied independently.

TABLE S8
QWEN3 JUDGMENTS ON THE AMAZON–GOOGLE TEST PAIRS.

Input	TP	FP	FN	Recall	F1
No peers	378	31	122	75.60	83.17
Peer decision	343	30	157	68.60	78.58
Decision + fields	279	34	221	55.80	68.63
Full explanation	287	29	213	57.40	70.34

500 matches and 500 non-matches per input; recall and F1 in percent.

Choice of shared content. DBLP–ACM and Amazon–Google compare four inputs with fixed Qwen2.5-7B peer outputs: no peers, a peer decision, that decision with schema-defined field comparisons, and the peer's full explanation. Reference entities define disjoint development and test sides, cross-side pairs are excluded, and samples are balanced by match label; development pair F1 selects an input per recipient, with ties broken by fewer input tokens. Table S9 reports every test alternative. On Amazon–Google, no peers has the highest development and test F1 for all three recipients, so selection avoids losses of 12.83, 3.90, and 3.20 pp against full explanations without improving on no peers; on DBLP–ACM development selection picks the peer decision for Qwen3 and no peers for gpt-oss and Gemma, and the gaps to the best tested input are 0.26, 0.12, and 0.85 pp, which describe this near-ceiling test set rather than a reliable ranking.

TABLE S9
SHARED CONTENT AND TEST PAIR F1 (%).

Input	Qwen3	gpt-oss	Gemma
DBLP–ACM (800 test pairs)			
No peers	99.12	99.00	98.52
Peer decision	98.87	99.12	99.37
Decision + fields	97.04	98.23	99.25
Full explanation	98.48	96.93	99.25
Amazon–Google (1,000 test pairs)			
No peers	83.17	91.02	91.63
Peer decision	78.58	90.56	90.27
Decision + fields	68.63	86.47	89.29
Full explanation	70.34	87.12	88.43

D. Stronger Peers

The source comparison replaces the Qwen2.5-7B peer responses by Qwen3-32B responses on the same 480 tasks and three positions, with the recipient's own no-peer response hidden. Qwen3-32B peers score 96.39, 77.78, 60.83, and 42.50% on GSM8K, StrategyQA, MMLU-Pro, and GPQA, against 91.94, 64.44, 55.56, and 31.11% for Qwen2.5-7B and 73.61, 54.72, 39.17, and 25.56% for Llama-3-8B; the two source conditions share the recipient model, prompts, and decoding settings but differ in peer text, random seeds, and answer extraction, so a source change alters content and style as well as accuracy. Using the no-peer response collected with the Llama-3-8B peer set as the baseline instead of the one collected with the Qwen2.5-7B set changes Qwen3's full and answers-only effects to $+0.62$ and $+5.00$ pp and gpt-oss's to $+5.07$ and $+8.96$ pp, with the same directions and Holm decisions. Table S10 gives the effects with Qwen3 peers and Table S11 the change from Qwen2.5 peers with its recovered and lost components; Recovered and Lost are the shares of all paired observations that change from wrong to correct and from correct to wrong, respectively, rather than rates conditional on initial correctness. Thus each change equals ΔRecovered $-$ ΔLost up to rounding, and these six changes with the three answers-minus-full changes ($+2.08$, -1.46, -1.39 pp) form one Holm family.

TABLE S10
SHARING EFFECTS WITH QWEN3-32B PEERS (PP).

Recipient	Contrast	Effect	95% CI
Qwen3	Full − none	$+2.36$	$[-0.14, 4.86]$
Qwen3	Answers − none	$+6.74$	$[4.31, 9.17]$
Qwen3	Answers − full	$+4.38$	$[2.22, 6.60]$
gpt-oss	Full − none	$+4.72$	$[2.01, 7.50]$
gpt-oss	Answers − none	$+8.61$	$[6.11, 11.11]$
gpt-oss	Answers − full	$+3.89$	$[2.08, 5.76]$
Gemma	Full − none	-0.07	$[-2.64, 2.50]$
Gemma	Answers − none	-0.69	$[-2.99, 1.53]$
Gemma	Answers − full	-0.62	$[-2.64, 1.39]$

480 tasks, three positions; one nine-contrast Holm family.

TABLE S11
CHANGE OF THE SHARING EFFECT WHEN QWEN3-32B PEERS REPLACE QWEN2.5-7B PEERS (PP).

Recipient	Input	ΔRecovered	ΔLost	Change	95% CI
Qwen3	Full	-2.30	-3.96	$+1.67$	$[-1.46, 4.79]$
	Answers	$+1.19$	-2.57	$+3.75^*$	$[1.32, 6.25]$
gpt-oss	Full	$+2.08$	-2.15	$+4.24^*$	$[1.39, 7.01]$
	Answers	$+1.94$	-0.84	$+2.78$	$[0.62, 4.93]$
Gemma	Full	$+1.11$	-2.64	$+3.75^*$	$[1.11, 6.32]$
	Answers	$+0.07$	-2.29	$+2.36$	$[-0.07, 4.72]$

*: passes the nine-change Holm family.

A weaker recipient with stronger peers. With Qwen3-32B peers, a Qwen2.5-7B recipient on the same 480 tasks reaches 68.33% with answers only and 71.81% with full responses, from 57.43% without peers (Table S12); the peers' pooled accuracy, 69.38%, exceeds the recipient's no-peer accuracy on every benchmark (gap 11.94 pp, 95% CI 8.54–15.49). Answers minus no peers ($+10.90$ pp, 95% CI 8.06–13.82) and full minus no peers ($+14.37$ pp, 10.69–18.06) pass the three-comparison Holm family that also contains the pooled answers-minus-full contrast. Answers-only recovers 17.36 pp of initial errors and loses 6.46 pp of initial successes; full responses recover 23.47 pp and lose 9.10 pp, so the additional corrections outweigh the additional regressions. With Qwen2.5 peers the same recipient instead favors answers alone by $+2.57$ pp (0.21–4.93), and the change between sources is -6.04 pp (-9.44 to -2.57), a descriptive contrast that jointly resamples the four source–format conditions. Mean input and output tokens per call are 199/430 with no peers, 240/416 with answers, and 1,120/467 with full responses; truncation rates are 2.50–2.71% and truncated outputs are scored as returned. The no-peer responses were generated separately with byte-identical prompts and settings, and re-scoring them agrees on all 1,440 outcomes.

S6. Q2: CANDIDATE FLOW AND FINAL DECISION

With the lenient extractor and the execution vote, the BIRD debates split by protocol: on 1,531 of the 1,534 BIRD development questions, Du-MAD's last-round accuracy with peers minus without is -2.30 pp, a loss with both backbones that survives the least favourable re-reading of replies that hold several queries. On the 198 sampled questions its real peers do worse than placebo peers, while in MAD-M^2 the direction depends on the backbone. QA traces distinguish lost candidates from unused ones; matching traces connect changed judgments to final group relations.

TABLE S12
QWEN2.5-7B RECIPIENT WITH QWEN3-32B PEERS: ACCURACY (%) AND ANSWERS MINUS FULL (PP).

Benchmark	No peers	Answers	Full	Answers − full [95% CI]
GSM8K	86.67	96.94	95.00	$+1.94\ [-0.83, 4.72]$
StrategyQA	60.28	77.50	76.11	$+1.39\ [-3.33, 6.39]$
MMLU-Pro	51.67	62.78	66.94	$-4.17\ [-10.01, 1.94]$
GPQA	31.11	36.11	49.17	$-13.06\ [-21.11, -5.28]$
All	57.43	68.33	71.81	$-3.47\ [-6.46, -0.56]$

1,440 outcomes per condition (three positions per task).

A. Candidate Flow by Protocol and Backbone

Table S13 expands the availability gaps in Section VI-A. Each row covers 198 BIRD questions with two repeats. C_0, C_T, C_D, Y have the definitions in Eq. 3; Y_x uses lenient extraction and execution voting. Loss and recovery are unconditional shares of runs whose pool respectively loses all correct candidates or acquires one. For BCCS, Y is the baseline terminal vote, while its actual returned-answer accuracies are 9.60% and 23.99%; stopping-dependent returns are not substituted into this flow identity.

TABLE S13
CANDIDATE FLOW AND BASELINE VOTE ACCURACY BY PROTOCOL–BACKBONE PAIR (%).

Protocol	Backbone	C_0	C_T	C_D	Y	Loss	Recovery	Y_x
Du-MAD	Llama-3-8B	23.99	23.99	0.00	0.00	1.52	1.52	22.98
	Qwen2.5-7B	36.87	35.61	35.61	34.09	3.28	2.02	35.35
MAD-M^2	Llama-3-8B	28.28	27.02	26.77	15.40	4.04	2.78	22.47
	Qwen2.5-7B	38.64	37.37	37.37	29.80	5.05	3.79	33.59
BCCS	Llama-3-8B	32.58	29.80	19.44	16.16	3.54	0.76	24.75
	Qwen2.5-7B	48.23	43.43	39.39	31.31	7.58	2.78	35.61
Mean		34.76	32.87	26.43	21.13	4.17	2.27	29.12

B. Per-Agent Transitions and Answer-Only Peers

The transitions below extend Section VI-D. On 1,531 questions, per-agent loss and recovery are shares of all initial-to-terminal agent trajectories; conditional rates instead divide by initially correct or initially wrong answers. These are distinct from pool-level availability and vote-level effects.

TABLE S14
PEER EFFECTS ON PER-AGENT TRANSITIONS, 1,531 BIRD QUESTIONS (PP).

Protocol	Gross loss	Gross recovery	Net	Conditional loss	Conditional recovery
Du-MAD	-1.12	-2.84	-1.73	-3.26	-4.33
MAD-M^2	-1.58	$+1.71$	$+3.29$	-7.22	$+2.25$
Mean	-1.35	-0.57	$+0.78$	-5.24	-1.04

The answer-only comparison uses 198 questions, two repeats, both protocols, and both backbones in each of three conditions: full responses, extracted SQL, and no peers. All 4,752 runs complete. The peer SQL is recovered using protocol extraction with lenient fallback; an unavailable answer contributes an empty peer answer rather than its explanation. Table S15 reports conditional loss and recovery. Their different denominators mean that recovery minus loss is not the total accuracy effect. The vote-level comparisons in Table VI subtract initial accuracy before contrasting conditions.

TABLE S15
CONDITIONAL PER-AGENT LOSS AND RECOVERY UNDER THREE PEER-CONTENT CONDITIONS (%).

		Full responses		Answers only		No peers	
Protocol	Backbone	Loss	Recovery	Loss	Recovery	Loss	Recovery
Du-MAD	Llama-3-8B	8.7	1.4	5.7	2.9	16.1	10.7
	Qwen2.5-7B	4.8	5.0	3.5	3.5	4.4	4.9
MAD-M^2	Llama-3-8B	51.9	9.5	65.3	7.5	56.5	7.4
	Qwen2.5-7B	23.7	11.1	59.9	5.0	33.1	12.0

The answers-only minus full-response peer interaction effects are $+0.38$ pp for Du-MAD and -0.88 pp for MAD-M^2, neither detected. The MAD-M^2/Qwen2.5-7B effect is -5.05 pp, detected. Rendering extracted peers in the expected answer format also raises Du-MAD/Llama-3-8B's returned accuracy under protocol extraction from 0.25% to 23.74%,

while its leniently read terminal vote changes from 21.46% to 23.74%. This gain therefore mixes content with format transfer; the body compares lenient extraction and execution voting on both sides.

C. Joint Output Changes and Tie Sensitivity

The extractor–vote interaction in Table IV crosses extraction and voting at the last round. For Du-MAD and MAD-M^2, its first-, last-, and random-tie estimates are $+0.25$, $+0.32$, and $+0.63$ pp, none detected. Including BCCS gives $+0.97$, $+0.76$, and $+1.39$ pp, all detected as fixed-trace comparisons. The first-tie joint change corrects 141 and regresses two of the 1,584 Du-MAD and MAD-M^2 runs. Changing the protocol vote's tie rule to last-agent ties instead corrects 43 and regresses 35, illustrating how a small net change can conceal different decisions.

TABLE S16
PEER INTERACTION EFFECTS UNDER ALTERNATIVE TIE RULES (PP).

Questions	Protocols and output rule	First	Last	Random
198	Du-MAD, MAD-M^2: protocol extractor, exact-match vote	$+3.03$	$+1.64$	$+1.07$
	Du-MAD, MAD-M^2: lenient extractor, execution vote	-1.96	-1.14	-2.15
	Du-MAD: lenient extractor, execution vote	-3.91	-3.91	-4.29
	MAD-M^2: lenient extractor, execution vote	0.00	$+1.64$	0.00
1,531	Du-MAD, MAD-M^2: lenient extractor, execution vote	-1.37	-0.79	-1.05
	Du-MAD: lenient extractor, execution vote	-2.74	-2.73	-2.68
	MAD-M^2: lenient extractor, execution vote	0.00	$+1.14$	$+0.57$

D. Reading Debate Replies

The per-class values behind Table IV and Fig. 4, including the four-configuration and six-configuration means under each extractor and vote and the BCCS divergence shares, are the extractor, vote, round, and peer contrasts on the BIRD debates in Table S1 (rows R4–R11). On matching, Du-MAD/Llama-3-8B's extractor finds a label in 23% of Settlements and 48% of WDC initial replies, whereas the lenient extractor finds one in 99% and agrees with 98.0–98.7% of the labels the protocol's extractor finds.

E. Controls on the Four Du-MAD and MAD-M^2 Configurations

Table S17 gives the contrasts behind Table V and the controls in Section VI-C, all read with the lenient extractor and the execution vote and paired per question. With 198 questions, the four-configuration design detects an effect of 2.1–2.8 pp at 80% power and a single configuration one of 3.6–7.8 pp; on the 1,531 questions the four-configuration design detects 1.25–1.35 pp.

The table gives two measures. The last-round difference is the last-round accuracy with peers minus without; the peer interaction effect subtracts the same difference taken on the initial answers.

The initial answers precede any exchange, and their prompts are byte-identical between the two conditions in every paired run, yet the replies and their accuracy still differ. MAD-M^2 samples at temperature 1 without a fixed seed, so the two conditions draw independent initial replies, and initial-answer correctness agrees in 80–88% of runs. Du-MAD decodes greedily, but greedy decoding on a shared server is not exactly reproducible, so only 25–38% of its initial replies are identical across the conditions, and initial-answer correctness agrees in 94.7–98.5% of runs. The initial-answer difference is therefore noise with zero expectation; on the 1,531 questions it is $+0.50$ pp over the four configurations, not detected, and positive in every configuration.

On the 1,531 questions the Du-MAD loss is detected on both measures and with each backbone. For MAD-M^2, the Llama-3-8B configuration gains and the Qwen2.5-7B configuration loses on the last-round difference, both detected, and only the Qwen2.5-7B configuration is detected on the peer interaction effect; the four-configuration last-round difference is not detected, whereas its peer interaction effect is. On the 198 questions the four-configuration last-round difference is -0.69 pp, not detected; the four-configuration peer interaction effect, -1.96 pp, has a database interval that excludes zero and a question interval whose upper end is exactly zero, and Du-MAD's two-backbone peer interaction effect is -3.91 pp, detected.

By difficulty, the simple questions carry a detected loss and the moderate and challenging ones are not detected, with no stratum showing a gain. Leaving out any one database keeps the 1,531-question effect between -1.70 to -1.17 pp, with both intervals excluding zero in every case; on the 198 questions the range is -2.50 to -1.66 pp, with the database interval excluding zero in all 11 cases and the question interval in 3.

Placebo peers are the replies of other questions from the same database, rendered in the protocol's own peer format; their round-1 prompts match the real ones in length (token ratio 1.01–1.02 per question). Real minus placebo on the change from the initial answers to the last round is -5.30 and -4.80 pp for Du-MAD with Llama-3-8B and Qwen2.5-7B,

both detected, and $+2.53$ and $+5.30$ pp for MAD-M^2, neither detected; placebo minus no peers is not detected in any configuration. In Du-MAD, real peers do worse than length- and format-matched replies to other questions, and the four-configuration mean, -0.57 pp, averages the two protocols' opposite signs.

At temperature 0.7 the share of Du-MAD runs whose initial queries return two or more distinct execution results is 32.1% (Llama-3-8B) and 35.6% (Qwen2.5-7B), against 4.0% and 7.6% when greedy, and the peer interaction effect is -2.53 pp with Llama-3-8B and -5.05 pp with Qwen2.5-7B.

The unified answer format asks every protocol for a fenced SQL block; it raises the final-answer accuracy under the protocol's own extractor by $+22.47$ pp for Du-MAD/Llama-3-8B (0% to 22.5%), -3.79 for Du-MAD/Qwen2.5-7B, $+4.80$ for MAD-M^2/Llama-3-8B, and -2.53 for MAD-M^2/Qwen2.5-7B, and the share of initial replies its own extractor finds by 100, 29, 9, and 0.3 pp.

TABLE S17
CONTROLS ON THE FOUR DU-MAD AND MAD-M^2 CONFIGURATIONS, LLAMA-3-8B AND QWEN2.5-7B AGENTS (PP).

Contrast	Est.	CI (questions)	CI (databases)
Peer interaction effect, remaining BIRD development questions			
Δ_{peer}, 1,333 remaining questions	-1.30	$[-2.25, -0.35]$	$[-2.33, -0.34]$
Δ_{peer}, all 1,531 questions	-1.37	$[-2.25, -0.51]$	$[-2.32, -0.43]$
Du-MAD, both backbones	-2.74	$[-3.77, -1.73]$	$[-3.46, -2.01]$
MAD-M^2, both backbones	0.00	$[-1.36, +1.34]$	$[-1.50, +1.46]$
Du-MAD / Llama-3-8B	-3.14	$[-4.70, -1.57]$	$[-4.43, -1.78]$
Du-MAD / Qwen2.5-7B	-2.35	$[-3.66, -1.05]$	$[-3.59, -1.19]$
MAD-M^2 / Llama-3-8B	$+1.89$	$[-0.26, +3.98]$	$[-0.41, +4.41]$
MAD-M^2 / Qwen2.5-7B	-1.89	$[-3.56, -0.20]$	$[-3.34, -0.69]$
simple (924 questions)	-1.80	$[-2.96, -0.64]$	$[-2.93, -0.66]$
moderate (463)	-0.84	$[-2.38, +0.73]$	$[-2.33, +0.54]$
challenging (144)	-0.35	$[-2.95, +2.17]$	$[-4.17, +3.56]$
Last-round difference, last round with − without peers			
all 1,531 questions	-0.87	$[-1.63, -0.13]$	$[-1.76, +0.18]$
Du-MAD, both backbones	-2.30	$[-3.36, -1.31]$	$[-3.18, -1.39]$
MAD-M^2, both backbones	$+0.56$	$[-0.52, +1.62]$	$[-0.77, +2.10]$
Du-MAD / Llama-3-8B	-2.97	$[-4.54, -1.47]$	$[-4.16, -1.71]$
Du-MAD / Qwen2.5-7B	-1.63	$[-2.91, -0.36]$	$[-2.85, -0.36]$
MAD-M^2 / Llama-3-8B	$+2.51$	$[+0.78, +4.25]$	$[+0.58, +4.70]$
MAD-M^2 / Qwen2.5-7B	-1.40	$[-2.78, -0.07]$	$[-2.47, -0.31]$
initial answers, with − without	$+0.50$	$[-0.03, +1.04]$	$[-0.05, +1.11]$
198 questions, four configurations	-0.69	$[-2.59, +1.20]$	$[-1.93, +0.57]$
Du-MAD, both backbones, peer interaction effect	-3.91	$[-6.69, -1.39]$	$[-5.90, -2.10]$
Placebo peers			
real − placebo, four configurations	-0.57	$[-2.97, +1.77]$	$[-2.81, +1.72]$
Du-MAD / Llama-3-8B	-5.30	$[-8.59, -2.53]$	$[-7.81, -2.82]$
Du-MAD / Qwen2.5-7B	-4.80	$[-8.59, -1.52]$	$[-7.87, -1.77]$
MAD-M^2 / Llama-3-8B	$+2.53$	$[-2.78, +7.83]$	$[-2.03, +6.75]$
MAD-M^2 / Qwen2.5-7B	$+5.30$	$[0.00, +10.61]$	$[-1.00, +10.80]$
placebo − no peers, four configurations	-1.26	$[-3.79, +1.26]$	$[-3.35, +1.01]$
Du-MAD / Llama-3-8B	0.00	$[-4.55, +4.80]$	$[-4.21, +4.50]$
Du-MAD / Qwen2.5-7B	$+2.78$	$[-0.76, +6.57]$	$[-1.30, +6.70]$
MAD-M^2 / Llama-3-8B	-4.55	$[-9.85, +0.51]$	$[-9.38, +0.75]$
MAD-M^2 / Qwen2.5-7B	-3.28	$[-9.09, +2.53]$	$[-8.04, +1.73]$
Temperature 0.7			
Δ_{peer}, Du-MAD	-3.79	$[-7.07, -0.51]$	$[-7.13, -0.37]$
Δ_{peer}, initial answers disagree	-3.37	$[-8.54, +1.57]$	$[-7.60, +1.33]$
diverse share, 0.7 − greedy	$+28.03$	$[23.99, 32.32]$	$[23.23, 33.09]$
Unified format			
final-answer accuracy, protocol's own extractor	$+5.24$	$[3.09, 7.39]$	$[2.95, 7.73]$
Δ_{peer} change	-0.06	$[-2.97, +2.90]$	$[-3.19, +3.22]$

Lenient extractor, execution vote, ties to the first agent; 198 questions unless stated; two repeats.

F. Sensitivity to Extractor Errors

Table S18 re-estimates the 1,531-question effects of Table S17 under readings that bound what errors of the lenient extractor could change. A recovered query is one that the protocol's own extractor misses and the lenient extractor finds. The lenient extractor takes the last candidate query in a reply, so it can return the wrong query only from a reply that offers a choice: a recovered query is *ambiguous* when its reply holds more than one candidate with different execution outcomes (a candidate that fails to execute counts as no vote). Ambiguous recovered replies are at most 2.4% of the initial replies and at most 0.4% of the last-round replies in the Du-MAD configurations. At the last round without peers, none of the 146 recovered queries whose removal alone changes a Du-MAD vote is ambiguous; with peers, 62 of the 64 runs with such a query hold only unambiguous ones.

Own extractor only scores the queries the protocol's extractor finds, with the execution vote, which drops every recovered query at once; Du-MAD/Llama-3-8B's extractor finds no query, so under this reading the configuration contributes zero to both conditions. Only the Qwen2.5-7B configuration contributes a nonzero effect to the two-backbone mean.

The *joint worst case* reassigns every ambiguous recovered reply, jointly across the agents of a run, to the candidate outcome least favourable to the estimate and re-runs the execution vote: with peers, correct at the last round and wrong in the initial answers where possible; without peers, the reverse; both directions where the estimate is zero. The first of its two columns uses the candidates the lenient extractor's own detection can return; the second uses a broader set that also takes every fenced block and every bare SELECT or WITH line.

Under the extractor's own candidates the Du-MAD loss stays detected on both measures and with each backbone, with 8 last-round votes and 28 votes in all reassigned over the two configurations; under the broader set the last-round loss stays detected and the peer interaction effect loses detection over databases. MAD-M^2/Llama-3-8B's last-round gain and MAD-M^2/Qwen2.5-7B's loss stay detected under every reading. The four-configuration peer interaction effect is not robust: it falls to -0.35 pp under the own-extractor reading and to -0.87 pp under the joint worst case, neither detected. The joint worst case covers an extraction that takes the wrong one of several candidates; a reply with a single candidate that is not the agent's intended final answer is covered by the audits (Section S2-D).

The second audit is a stratified sample of the 13,027 recovered replies that decide an execution vote in the Du-MAD runs on the 1,531 questions, alone or jointly with the other recovered queries of the run: 12, 13, 12, and 13 replies drawn uniformly within the four strata (Llama-3-8B with and without peers, 5,840 and 6,090 replies; Qwen2.5-7B with and without peers, 559 and 538). The annotator accepted all 50 as the intended final answer: 12/12, 13/13, 12/12, and 13/13, with Wilson 95% lower bounds of 75.8%, 77.2%, 75.8%, and 77.2% per stratum and 92.9% for the 50 sampled replies taken together (not a population-weighted estimate). Treating rejected queries as no answer therefore changes no sampled vote.

As a sensitivity at an assumed error rate rather than a bound, flipping the votes that a single recovered query decides, whether or not its reply offers an alternative, at the audited rates of 1 of 50 and 12 of 450 in 1,000 random draws moves the Du-MAD two-backbone estimates by less than 0.1 pp.

TABLE S18
SENSITIVITY OF THE 1,531-QUESTION EFFECTS TO ERRORS OF THE LENIENT EXTRACTOR (PP).

Models	Measure	Lenient extractor	Protocol extractor	Joint worst case, extractor's candidates	Joint worst case, broad candidates
Du-MAD, both backbones	last-round	-2.30✓	-1.00✓ $[-1.65, -0.38]$ $[-1.53, -0.42]$	-2.17✓ $[-3.23, -1.18]$ $[-3.06, -1.29]$	-1.62✓ $[-2.65, -0.62]$ $[-2.72, -0.48]$
	Δ_{peer}	-2.74✓	-1.21✓ $[-1.94, -0.49]$ $[-1.82, -0.65]$	-2.29✓ $[-3.33, -1.27]$ $[-3.05, -1.53]$	-1.29 $[-2.37, -0.21]$ $[-2.58, +0.35]$
Du-MAD / Llama-3-8B	last-round	-2.97✓	0.00 (no query found)	-2.71✓ $[-4.31, -1.21]$ $[-3.90, -1.51]$	-1.60✓ $[-3.17, -0.07]$ $[-3.16, -0.03]$
	Δ_{peer}	-3.14✓	0.00 (no query found)	-2.22✓ $[-3.79, -0.65]$ $[-3.47, -0.92]$	-0.26 $[-1.96, +1.44]$ $[-2.23, +2.15]$
Du-MAD / Qwen2.5-7B	last-round	-1.63✓	-1.99✓ $[-3.30, -0.75]$ $[-3.06, -0.85]$	-1.63✓ (no vote changes)	-1.63✓ (no vote changes)
	Δ_{peer}	-2.35✓	-2.42✓ $[-3.89, -0.98]$ $[-3.64, -1.29]$	-2.35✓ (no vote changes)	-2.32✓ $[-3.63, -1.01]$ $[-3.55, -1.17]$
MAD-M^2, both backbones	last-round	$+0.56$	$+0.93$ $[-0.15, +1.99]$ $[-0.42, +2.47]$	$+0.51$ $[-0.57, +1.57]$ $[-0.81, +2.04]$	$+0.47$ $[-0.62, +1.53]$ $[-0.86, +2.01]$
	Δ_{peer}	0.00	$+0.51$ $[-0.82, +1.85]$ $[-0.93, +1.79]$	$+0.54$ $[-0.80, +1.86]$ $[-1.03, +2.12]$	$+0.65$ $[-0.70, +1.99]$ $[-0.89, +2.22]$
	other direction			-0.31 $[-1.67, +1.03]$ $[-1.78, +1.07]$	-0.44 $[-1.81, +0.90]$ $[-1.96, +0.96]$
MAD-M^2 / Llama-3-8B	last-round	$+2.51$✓	$+3.27$✓ $[+1.57, +5.00]$ $[+1.31, +5.43]$	$+2.42$✓ $[+0.69, +4.15]$ $[+0.49, +4.58]$	$+2.35$✓ $[+0.62, +4.08]$ $[+0.41, +4.56]$
	Δ_{peer}	$+1.89$	$+2.97$✓ $[+0.82, +5.09]$ $[+0.85, +5.06]$	$+1.31$ $[-0.85, +3.40]$ $[-0.93, +3.68]$	$+1.05$ $[-1.11, +3.14]$ $[-1.27, +3.47]$
MAD-M^2 / Qwen2.5-7B	last-round	-1.40✓	-1.40✓ $[-2.78, -0.07]$ $[-2.47, -0.31]$	-1.40✓ (no vote changes)	-1.40✓ (no vote changes)
	Δ_{peer}	-1.89✓	-1.96✓ $[-3.59, -0.29]$ $[-3.36, -0.83]$	-1.89✓ (no vote changes)	-1.89✓ (no vote changes)
Pooled	last-round	-0.87	-0.03 $[-0.67, +0.58]$ $[-0.90, +0.98]$	-0.69 $[-1.44, +0.05]$ $[-1.59, +0.38]$	-0.39 $[-1.13, +0.34]$ $[-1.40, +0.74]$
	Δ_{peer}	-1.37✓	-0.35 $[-1.14, +0.42]$ $[-1.28, +0.51]$	-0.87 $[-1.75, 0.00]$ $[-1.91, +0.20]$	-0.32 $[-1.21, +0.59]$ $[-1.51, +0.97]$

✓: detected, both 95% intervals (questions; databases) exclude zero; lenient-extractor intervals in Table S17; joint worst case: every ambiguous recovered reply reassigned jointly to the outcome least favourable to the estimate.

G. Failure Classes and Accuracy Levels

Table S19 splits every wrong final answer on BIRD of the protocol matrix (Llama-3-8B and Qwen2.5-7B agents), with peers, into three exhaustive classes: a candidate-absence failure, where no reply in the run holds a correct query under the lenient extractor; a format failure, where a reply holds a correct query that the protocol's own extractor misses; and a selection failure, where the protocol's extractor finds a correct query but the vote, stop rule, aggregator, or return step does not return it. The candidate set is every agent's initial reply and last-round reply; using every round moves no class by more than 2 pp. For Du-MAD/Llama-3-8B every format failure delivers no answer at all, which is the 0/400 of Fig. 3; even with a perfect extractor, 74.5% of its runs hold no correct query. Table S20 gives the accuracy of the four Du-MAD and MAD-M^2 configurations with peers under the lenient extractor and the execution vote, and their final-answer accuracy under the protocol's own extractor. Across the 16 cross-family models the lenient-read accuracy ranges from 13.38% (gpt-oss-20b, Du-MAD, last round) to 59.09% (Qwen3.8-27B, Du-MAD, last round); Agentar-SQL-32B, Gemma-4-26B-A4B, Qwen3.6-35B-A3B, and Qwen3.8-27B exceed 50%.

TABLE S19
WHY A FINAL ANSWER ON BIRD IS WRONG, WITH PEERS (% OF 396 RUNS PER CONFIGURATION).

Cell	Final answer correct	Candidate absence	Format	Selection
Du-MAD / Llama-3-8B	0.0	74.5	25.5	0.0
Du-MAD / Qwen2.5-7B	34.1	61.1	0.0	4.8
MAD-M^2 / Llama-3-8B	15.4	68.9	0.8	14.9
MAD-M^2 / Qwen2.5-7B	29.8	57.6	0.0	12.6
BCCS / Llama-3-8B	9.6	66.7	12.1	11.6
BCCS / Qwen2.5-7B	24.0	49.0	4.3	22.7
MoA (roster)	39.1	37.1	0.0	23.7
FORD (roster)	28.5	65.9	0.0	5.6

Candidate absence: no reply holds a correct query; format: the extractor missed one; selection: one was read but not returned.

TABLE S20
ACCURACY OF THE FOUR DU-MAD AND MAD-M^2 CONFIGURATIONS WITH PEERS (%).

Cell	Initial answers	Last round	Final answer
Du-MAD / Llama-3-8B	23.23	22.98	0.0
MAD-M^2 / Llama-3-8B	23.99	22.47	15.4
Du-MAD / Qwen2.5-7B	35.86	35.35	34.1
MAD-M^2 / Qwen2.5-7B	32.83	33.59	29.8

Initial answers and last round: lenient extractor, execution vote; final answer: the protocol's own extractor and vote.

H. Grouping

The reversals, examples, and relation counts below support the grouping results of Section VI-F. *Reversals between pair and cluster F1.* Holding each grouping rule fixed, the entity-disjoint test matrix has 66 sharing configurations (three datasets, two recipients, 11 sharing configurations including random-peer draws), each evaluated with 15 rules. Of the 990 combinations, 655 improve both metrics, 272 lower both, 34 reverse, and 29 change pair F1 with unchanged cluster F1; 13 of the 66 configurations reverse under at least one rule, and treating changes below 0.10 pp as near zero leaves 11 reversals (Table S21). Pair-up/group-down reversals have a median cluster change of -0.27 pp (range -3.38 to -0.01); pair-down/group-up gains stay below 0.30 pp. Center clustering has no reversals; reversals occur only in link-propagating families. The counts describe correlated configurations with unequal rule-family menus.

The full-population Settlements example. Qwen2.5-32B decision sharing changes pair and cluster F1 by $+0.026$ and -3.053 pp. Twenty-three direct non-match judgments become matches (12 true, 11 false) with no reverse changes; under connected components, nine component merges add 147 group relations, 14 direct and 133 indirect, and all 133 indirect additions are false (Table S22). The largest merge joins four components into 15 records spanning five reference entities and contributes 78 false indirect pairs. A separate judgment collection on the same 1,494 records and 4,719 pairs changes cluster F1 by -0.55 pp; the two judgment collections are analyzed separately, and reference entities identify these errors retrospectively rather than as a label-free predictor.

In the Settlements example, 14 changed edges join previously separate components, nine incorrectly. Each incorrect bridge violates the cluster-F1 improvement condition; applied individually, these nine edges add 130 relations, all false. Across all 14 joining edges, the individual additions total 141 relations, while their joint closure adds 147. The largest single-edge addition has 20 relations, so the error is distributed across bridges. The second judgment collection on the same records changes ten judgments and adds 24 relations, including 22 false indirect relations, with pair F1 $+0.05$ pp

TABLE S21
SHARING-EFFECT REVERSALS BETWEEN PAIR F1 AND CLUSTER F1.

	Combinations	Pair ↑, group ↓	Pair ↓, group ↑	Near zero
By dataset				
WDC	330	2	11	2
Settlements	330	18	3	9
MusicBrainz	330	0	0	0
By grouping family				
Center	66	0	0	
Connected components	66	3	1	
Merge-center	66	2	1	
Disjoint-star	66	2	0	
Greedy correlation	396	6	9	
Markov	330	7	3	

TABLE S22
ADDED AND REMOVED GROUP RELATIONS ON FULL-POPULATION SETTLEMENTS.

Contrast (Qwen2.5-32B, test)	a_T	a_F	r_T	r_F	ΔF1 (pp)
Decisions − no peers	5	142	0	0	−3.05

a: added, r: removed relations; T, F: true, false (Section S2-E).

and cluster F1 -0.55 pp. This repeats the direction of the pair-versus-group discrepancy with a different magnitude. Across 66 test configurations, varying the grouping setting changes the sign of the sharing effect in 13.

I. Candidate Availability in QA Traces

Correct candidates lost during QA debate. Across the eight protocol variants in nine collections (Table S23; single-model runs use Llama-3-8B and Qwen2.5-7B, Du-MAD and MAD-M^2 with its objective filter also Mistral-7B, mixed-model runs the three models, and every variant covers GSM8K, StrategyQA, MMLU-Pro, and GPQA, with Du-MAD also on SQuAD2), traces exist for 10,295 of the 10,900 planned runs; 597 of the 605 missing traces belong to the two DoV-style collections. Among the 5,785 traced successes, 5,574 (96.4%) already contain a correct answer before interaction. Of the 7,044 runs that start with a correct candidate, 1,470 fail: 701 still hold one at final selection, 494 lose every correct candidate, and for 275 the availability cannot be determined, so the availability loss is 7.30% among the 6,769 evaluable endpoints (7.01–10.92% including the undetermined cases). Among 2,224 runs whose initial candidates are all wrong, 198 later contain a correct answer (8.90%) and 106 return one (4.77%). Loss conditions on an initially correct candidate and known terminal availability; Miss conditions on a correct terminal candidate and a scored final outcome, so the 701 retained-candidate failures are not its numerator: the primary Miss rate is 838/6,623 (12.65%), or 852/6,637 (12.84%) counting 14 unscorable final answers as failures.

TABLE S23
QA DEBATE TRACES: COVERAGE AND THE STAGE AT WHICH A CORRECT CANDIDATE IS LOST.

Variant	Planned	Traced	Missing	Lost / retained failures	Loss	Miss
Du-MAD	3,000	2,995	5	61 / 43	61/1,686	64/1,736
BCCS	800	800	0	26 / 101	26/534	107/518
DMAD	800	800	0	132 / 11	132/522	11/402
DoV-style debate, collection 1	800	501	299	25 / 5	25/341	6/333
DoV-style debate, collection 2	800	502	298	25 / 3	25/349	1/342
MAD-M^2, all peers	800	800	0	29 / 94	29/563	109/566
MAD-M^2, objective filter	1,200	1,200	0	43 / 130	43/758	169/778
Du-MAD, mixed models	1,500	1,497	3	81 / 119	81/1,152	145/1,107
MAD-M^2, objective filter, mixed models	1,200	1,200	0	72 / 195	72/864	226/841
All	10,900	10,295	605	494 / 701	494/6,769	838/6,623

Rows cover different tasks and model configurations; their scores are not paired comparisons.

S7. Q3: STAGE CONTRIBUTIONS AND SIMPLER REPLACEMENTS

Stage comparisons relate the frequency of changed results and the net gain on those tasks to the quality effect and cost of replacement. Table S24 and the replacement table in the body both report stage minus rule.

A. Extra Stages

Table S24 gives the counts behind Fig. 6: on how many tasks each stage changes the result, how those tasks split into gains, losses, and both wrong, and the stage's share of its pipeline's calls and tokens. A result changes when the two outputs fall in different scorer classes (SQL result sets; ComEM decision vectors); intervals resample databases for SQL and are stratified by dataset for ComEM.

TABLE S24
EXTRA STAGES: HOW OFTEN EACH CHANGES THE RESULT, AND ITS GAIN WHEN IT DOES (PP).

Stage → rule	Model, data	n	Changed % (n)	Gain / loss / both wrong	Gain if changed	Stage − rule [95% CI]	Calls / tokens
MAC-SQL Refiner → Decomposer	Qwen3-32B, BIRD	1,534	8.08 (124)	36 / 8 / 80	$+22.58$	$+1.83$ $[1.00, 2.70]$	10.9% / 6.1%
	Qwen2.5-72B, BIRD	1,534	10.43 (160)	63 / 10 / 87	$+33.12$	$+3.46$ $[2.06, 5.01]$	13.1% / 7.3%
	Qwen3-32B, Spider	1,034	0.77 (8)	4 / 0 / 4	—	$+0.39$ $[0.00, 0.84]$	0.6% / 0.4%
	Qwen2.5-72B, Spider	1,034	1.84 (19)	10 / 0 / 9	—	$+0.97$ $[0.00, 2.13]$	1.4% / 1.0%
CHESS Unit Tester → self-consistency	Qwen3-32B, BIRD	147	15.65 (23)	4 / 6 / 13	—	-1.36 $[-5.71, 2.88]$	37.1% / 23.0%
ComEM selector → ranker	Qwen3-32B, 8 datasets	3,200	26.66 (853)	466 / 300 / 87	$+19.46$	$+5.19$ $[3.53, 6.84]$	all
	Qwen2.5-7B, 8 datasets	3,200	30.53 (977)	456 / 414 / 107	$+4.30$	$+1.31$ $[-0.44, 3.06]$	all

—: fewer than 30 changed tasks.

B. The CHESS Unit Tester

The comparison boundary is the 20-candidate list after revision and before the Unit Tester begins; the rules use this list and its execution results, whereas the Unit Tester also reads the question, hint, schema, and its generated tests. The candidate lists are identical at the boundary. All 147 questions remain in the denominator. The Unit Tester is entered on 145 of them: on questions 81 and 690 a tool-choice response ends the agent early and CHESS returns a revised-candidate fallback (incorrect and correct, respectively), and on question 515 a 40,961-token test-generation request exceeds the context limit and the evaluation falls back without tests to an incorrect query. The returned branch therefore has 83 correct answers, or 82 if the three flagged outcomes are scored as failures. Table S25 gives the branch counts, the paired contrasts, and their sensitivity to scope and weighting; no contrast passes the three-comparison Holm adjustment, and the few discordant outcomes limit precision, so the comparisons do not establish equivalence. Self-consistency, Valid, and Nonempty agree with the Unit Tester's execution result on 114, 115, and 116 questions, while their SQL strings agree on only 94, 10, and 8, because different tie-selected queries can return the same answer; self-consistency and Valid have identical per-question correctness.

TABLE S25
CHESS: BRANCHES, CONTRASTS, AND SENSITIVITY ON 147 BIRD QUESTIONS.

Branch	Correct	Acc. (%)	Stage calls
Unit Tester (returned)	83	56.46	2,599
Self-consistency	85	57.82	0
Valid	85	57.82	0
Nonempty	88	59.86	0
Any correct candidate	105	71.43	—
Unit Tester − rule (pp)	**Self-cons.**	**Valid**	**Nonempty**
All 147, question weights	-1.36	-1.36	-3.40
95% CI	$[-5.71, 2.88]$	$[-5.71, 2.88]$	$[-7.69, 0.00]$
Flagged scored as failures	-2.04	-2.04	-4.08
Entered only (145)	-1.38	-1.38	-3.45
All 147, database weights	-1.85	-1.85	-4.43

Intervals over databases; any-correct is an oracle reference.

TABLE S26
CHESS: SHARED PREFIX AND THE UNIT TESTER'S INCREMENT.

Work	Calls	Input tokens	Output tokens
Shared prefix	4,402	38,073,165	1,829,855
Unit Tester	2,599	9,780,449	2,108,548
Execution vote	0	0	0

All 147 questions; prefix: retrieval, generation, and revision.

The Unit Tester comprises 437 tool-choice, 102 test-generation, and 2,060 evaluation calls, 37.1% of the 7,001 calls and 53.5% of the completion tokens (Table S26). The execution vote's candidate executions take at most 3.53 s for one question. The full workflow's median wall time per question is 112 s (maximum 287 s) and its server occupancy about 5.2 GPU-hours on three Qwen3 replicas including loading and idle time, so call reductions are not estimates of end-to-end latency or cost reductions.

C. Execution Filters on the Qwen2.5 Pools

Table S27 gives every Qwen2.5 generation-minus-voting contrast behind Table VIII. Valid gains 4.57 and 5.42 pp over Unfiltered on BIRD and 0.77 and 1.35 pp on Spider with four and six candidates, without losing a correct selection or adding a call. A 72B recipient on the four-candidate pool exceeds Valid on BIRD by 3.46 pp and differs from the 32B recipient by -0.22 pp on BIRD and $+0.97$ pp on Spider. Paired outcomes separate operations with similar scores: Unfiltered six-candidate Spider voting and Qwen2.5-32B generation score 74.37% and 74.27%, yet succeed alone on 114 and 111 runs, so 7.25% of the 3,102 outcomes differ.

TABLE S27
QWEN2.5 GENERATION MINUS VOTING (PP).

	BIRD		Spider	
Recipient / candidates	Δ	95% CI	Δ	95% CI
Generation minus Unfiltered				
7B / 4	$+3.37$	$[0.06, 6.16]$	-1.71	$[-3.26, -0.34]$
32B / 4	$+8.24$	$[5.98, 10.80]$	$+0.87$	$[-1.80, 3.58]$
7B / 6	$+3.57$	$[-0.02, 7.47]$	-3.16	$[-5.21, -1.40]$
32B / 6	$+8.70$	$[6.41, 11.20]$	-0.10	$[-2.38, 2.07]$
Generation minus Valid				
7B / 4	-1.20	$[-4.48, 1.82]$	-2.48	$[-4.12, -1.02]$
32B / 4	$+3.67$	$[1.92, 5.44]$	$+0.10$	$[-2.64, 2.87]$
7B / 6	-1.85	$[-5.01, 1.26]$	-4.51	$[-6.09, -3.03]$
32B / 6	$+3.29$	$[1.63, 5.42]$	-1.45	$[-3.35, 0.35]$
Generation minus Nonempty				
7B / 4	-3.87	$[-7.32, -1.22]$	-2.97	$[-4.70, -1.39]$
32B / 4	$+1.00$	$[-0.66, 2.76]$	-0.39	$[-3.19, 2.20]$
7B / 6	-5.18	$[-8.31, -2.21]$	-5.00	$[-6.96, -2.76]$
32B / 6	-0.04	$[-1.57, 1.42]$	-1.93	$[-4.10, 0.16]$

Question weights; intervals over databases; three generation runs.

D. Execution Cost, Empty Winners, and Test-Suite Scoring

Table S28 places the execution votes of the body next to the LLM stage each replaces. SQL seconds are logged single-threaded SQLite times per question (medians in the millisecond range; 5-second timeouts set the tail), and GPU-seconds use the throughput estimates defined below for local Qwen3-32B calls (the Llama-3-8B and Qwen2.5-7B debates have no such estimate, and DeepSeek-V4-Pro is hosted). Table S29 counts execution-vote winners that return an empty result: on BIRD no reference answer is empty, so every such winner is wrong, and a non-empty preference removes about half of them. Table S30 re-scores every Spider candidate and output on the test-suite databases of Zhong et al. (25–60 per question, 38.8 on average), a query counting as correct only when its result set equals the reference's on every database; 20.5–24.8% of the groups that the vote treats as equal on the development database split on at least one test-suite database, and 10.3–10.7% of Valid winners are correct on the single instance but not on the suite.

GPU-time estimation. We estimate LLM work on the H200 host from logged input and output token counts, p and c, as $p/R_p + c/R_d$, where R_p and R_d are prefill and decoding rates. Rates are fitted to per-call latency using calls with no overlapping request on the same server; the point estimate uses the model's median single-stream fit. Qwen3-32B decoding rates are 49.97–51.70 tokens/s, and Qwen2.5-72B-AWQ rates are 40.21–40.56 tokens/s. Sensitivity ranges use the fastest observed rates, including batched throughput where available, and the slowest single-stream fits. Prefill rates are less precisely identified because prompt counts include cached prefixes. The estimate assumes non-overlapping prefill and decoding and measures LLM work, not reserved GPU time or end-to-end latency. In Table VII, GPU share divides the replaced stage's estimated time by that of the stage and its preceding LLM calls. ComEM's selector is the only LLM call in this accounting; its Flan-T5 ranker time is measured separately.

E. The Grouping Menu

The settings below are the menu from which Section VII-C selects. The menu has 15 settings in six families: center clustering; merge-center clustering, which also joins groups when a link touches a center; connected components; disjoint stars, which group each highest-degree unassigned record with its unassigned neighbors; greedy correlation

TABLE S28
EXECUTION VOTES NEXT TO THE LLM STAGE EACH REPLACES, PER QUESTION.

Setting	Method	EX %	LLM calls	LLM tokens	GPU-s	SQL s
BIRD, four-model pool (1,534)	Valid / Nonempty vote	52.15 / 54.82	0	0	0	0.104
	Qwen3-32B selector	55.93	1	2,494	27.8	0
	Qwen3-32B generation	58.28	1	2,147	21.1	0
	DeepSeek-V4-Pro selector / generation	56.84 / 62.97	1	3,592 / 3,615	—	0
Spider, four-model pool (1,034)	Valid / Nonempty vote	73.31 / 73.79	0	0	0	0.018
	Qwen3-32B selector / generation	74.66 / 73.98	1	1,820 / 1,360	24.7 / 15.9	0
CHESS, Qwen3-32B (147)	Valid / Nonempty vote	57.82 / 59.86	0	0	0	0.258
	Unit Tester	56.46	17.68	80,878	287.8	—
BIRD debates, Llama-3-8B or Qwen2.5-7B, with peers (198)	Execution vote over initial answers	23.2–39.6	0	0	—	0.02–0.19
	Later rounds and final answer	0.0–39.1	$+1$ to $+10.4$	$+1.8$k to $+21.0$k	—	0

Each question weighted equally; tokens in and out; GPU-s for one request at a time (1.5–26.9 when batched); SQL s is the mean single-threaded SQLite time per question.

TABLE S29
EXECUTION-VOTE WINNERS THAT RETURN AN EMPTY RESULT (%).

Setting	Rule	Of questions	Of wrong winners
BIRD pool, four candidates	Valid	12.65	26.47
	Nonempty	5.74	12.72
BIRD pool, six candidates	Valid	12.73	27.50
	Nonempty	4.24	9.88
Spider pool, four candidates	Valid	9.09	17.75
	Nonempty	5.51	7.38
Du-MAD / Llama-3-8B, initial answers	execution vote	13.51	29.6
Du-MAD / Qwen2.5-7B, initial answers	execution vote	10.86	26.6
MAD-M^2 / Llama-3-8B, initial answers	execution vote	14.65	30.3
MAD-M^2 / Qwen2.5-7B, initial answers	execution vote	14.90	29.1
BCCS / Llama-3-8B, initial answers	execution vote	17.30	28.7
BCCS / Qwen2.5-7B, initial answers	execution vote	17.55	32.6
MoA references	execution vote	16.16	27.7

BIRD has no empty reference answer; on Spider, 45 of the 94 empty Valid winners are correct.

clustering, which merges groups while their summed link weights exceed θ times the number of cross-group pairs, for $\theta \in \{0, 1/6, \ldots, 5/6\}$; and Markov clustering at inflation $\{1.4, 1.8, 2, 2.5, 3\}$.

Development cluster F1 selects a setting, with ties broken by family order and then by increasing parameter. Connected components is selected in 20 of 24 WDC configurations, greedy correlation in 23 of 24 Settlements configurations, and on MusicBrainz connected components in 13, Markov clustering in 10, and greedy correlation in 1. Mean test gaps to the best setting are 0.023, 0.028, and 0.069 pp on WDC, Settlements, and MusicBrainz, with maxima of 0.244, 0.185, and 0.205 pp.

The development sides hold 419 labeled entities (1,560 records) on Settlements, 254 (2,936) on WDC, and 2,031 (3,981) on MusicBrainz, and running all 15 settings takes 0.095–0.130, 0.027–0.052, and 0.124–0.171 s per configuration, timed around the grouping algorithms only.

Scoring cluster F1 on sampled development entities while keeping the full graph gives Settlements mean test gaps

TABLE S30
SPIDER ACCURACY ON THE SINGLE DEVELOPMENT DATABASE AND ON THE TEST-SUITE DATABASES.

System (four / six candidates)	Single	Suite	Drop	Contrast (pp)	Single	Suite	Detected
Unfiltered vote	72.44 / 74.37	62.28 / 64.02	10.2 / 10.3	Valid − Unfiltered, four / six	$+0.77$ / $+1.35$	$+0.58$ / $+0.97$	yes / yes
Valid vote	73.21 / 75.73	62.86 / 64.99	10.3 / 10.7	Nonempty − Valid, four / six	$+0.48$ / $+0.48$	$+0.68$ / $+0.48$	no / no
Nonempty vote	73.69 / 76.21	63.54 / 65.47	10.2 / 10.7	Qwen2.5 generation − Valid, four	-1.10	$+1.34$	no / no
Qwen2.5 generation	72.11 / 72.74	64.20 / 64.60	7.9 / 8.1	Qwen2.5 generation − Valid, six	-2.98	-0.39	yes / no
Qwen3-32B selector / generation, four	74.66 / 73.98	66.15 / 65.76	8.5 / 8.2	Qwen3-32B selector − Valid	$+1.45$	$+3.29$	no / no
DeepSeek-V4-Pro selector / generation, four	76.31 / 75.63	67.31 / 64.22	9.0 / 11.4	DeepSeek-V4-Pro selector − Valid	$+3.09$	$+4.45$	yes / yes
				DeepSeek-V4-Pro selector − Nonempty	$+2.61$	$+3.77$	no / yes
				DeepSeek-V4-Pro generation − selector	-0.68	-3.09	no / yes

Accuracy in %, each question weighted equally; detected: both intervals exclude zero, on the single instance / on the suite; Qwen2.5 generation: mean of the Qwen2.5-7B and 32B recipients, three runs each.

TABLE S31
SENSITIVITY OF CONCLUSIONS TO EXTRACTION, VOTING, AND EVALUATION LEVEL.

Baseline → alternative	Cases	Sign	Detect.	Stable
Same outcome, alternative output rules				
BIRD sharing effect	6	3	3	2
BIRD Δ_{peer}	5	2	2	1
BIRD final answer−initial-answer vote	7	4	6	1
EM, QA: final answer−initial answers	80	7	5	70
cases where the extractor finds every answer	27	0	0	27
Different operation or metric				
Return rule→majority, ties to first agent	40	8	7	27
ties to last agent / kept	40	3/2	4/3	33/35
Pair F1→cluster F1	990	34	–	927

of 4.009, 2.624, 1.244, 0.315, and 0.028 pp with 21, 84, 126, 210, and 419 development entities (200 replicates, 24 configurations; fixed center clustering gives 1.782 pp): cross-entity false merges become invisible under entity subsampling, as the false pairs of connected components fall from 200.2 at full size to 51.4 at half size and 1.5 at one tenth while recall stays near 0.96.

F. Return Rules on Fixed Answers

The QA replay uses 112 tasks selected by a fixed rule independent of outcomes. Each task has five fixed sets of three initial responses, from Llama-3-8B, Qwen2.5-7B, or Qwen2.5-32B agents alone and from mixed models; each agent receives the one peer reply whose perplexity is above the median, and three independently sampled updates from each fixed set yield 1,680 sets of post-update answers. Candidates are the three post-update answers. The original plurality counts unreadable answers as one answer category; the readable-only plurality removes them; both break ties by agent order. Task-level resampling that keeps the dependence between sets and repeats gives an accuracy difference of $+3.15$ pp (95% CI 1.90–4.58).

G. Sensitivity across Output Rules

The following sensitivity summary extends the extraction, voting, and grouping comparisons in Sections VI and VII.

Fix the extractor and the comparison of answers before comparing protocols. On BIRD with first-agent ties, changing the extractor and the vote moves the Du-MAD and MAD-M^2 peer interaction effect from $+3.03$ to -1.96 pp (Section VI-B). A comparison of consensus with voting on data tasks should apply one extractor and one comparison to both sides and report the tie rule. The tie rate bounds that effect: 34 pp for MoA (ties in 95% of its runs), at most 2.0 pp for the matching protocols (0.4–6%). Extending Debate-or-Vote's extractor finding [8], we count how often changing output rules changes a conclusion (Table S31). On BIRD the baseline is the protocol's final answer or its own extractor with exact-match voting, and the alternative uses lenient extraction with execution voting; on matching and QA both sides are re-read leniently. A case flips its sign (Sign) or whether its interval excludes zero (Detect.), or stays stable; the Du-MAD/Llama-3-8B configuration, whose extractor delivers nothing, is omitted from the first two rows. Alternative output rules change the conclusion in 24 of 98 eligible cases, concentrated where the protocol's extractor misses replies; all 27 cases whose extractor finds every answer stay stable.

S8. Q4: ADDITIONAL COMPUTATION AND CANDIDATE SUPPLY

With DeepSeek-V4-Pro both a selection and a generation call beat voting on BIRD, and the stronger backbone's generation gain, not its selector gain, is detectably larger than Qwen3-32B's; larger Qwen2.5 debate backbones do not change the peer interaction effect detectably.

A. Selection and Final-Answer Generation

Table S32 tests the two backbones against each other on the same questions and pool: the difference of their selector gains over Nonempty and of their generation gains over Nonempty, which cancels the shared baseline. On BIRD the selector difference is not detected under any weighting, the generation difference is detected under all three (each database weighted equally; each question weighted equally with database clusters; each question weighted equally with question resampling), and the generation-minus-selector gap is 4.39 pp wider for DeepSeek-V4-Pro.

Table S33 gives the cross-family contrasts behind Fig. 7 with their Holm decisions. The Gemma–Spider contrasts against Valid and Nonempty pass a separate three-contrast family but not the joint one.

Tables S34–S36 report LLM selection with full inputs by generation repeat and pool. Both repeats use identical prompts, settings, and budgets. The execution-summary selector compared in Table S34 uses reasoning.

TABLE S32
DEEPSEEK-V4-PRO MINUS QWEN3-32B: GAINS OVER NONEMPTY ON THE SAME QUESTIONS (PP).

Benchmark	Gain	Difference	95% CI (databases)
BIRD	selector	$+0.87$	$[-0.38, +2.10]$
BIRD	generation	$+5.26$	$[3.18, 7.40]$
BIRD	generation − selector gap	$+4.39$	$[2.18, 7.07]$
Spider	selector	$+0.85$	$[-0.49, +2.08]$
Spider	generation	$+1.40$	$[-0.31, +3.05]$
Spider	generation − selector gap	$+0.55$	$[-1.33, +2.54]$

Each database weighted equally; one generation per question for both models.

Table S37 decomposes candidate availability. Table S38 reports Qwen3-32B quality and incremental cost after candidate acquisition and execution. Generation and full-input selection average two repeats; the main text's 57.88% uses the first. The execution-summary selector in this cost table uses one response per question without reasoning.

Within-pool comparisons use all 1,534 BIRD and 1,034 Spider questions; cross-pool comparisons use the 1,532 BIRD questions with scored candidates in both pools, on all 1,534 the four-model pool's availability is 63.30%.

Across pools, Nonempty's BIRD accuracy rises by 3.08 pp (95% CI 1.56–4.59), LLM selection with full inputs by 1.97 pp (0.08–4.06), and generation changes by -0.34 pp (-2.33 to 1.55).

Within each BIRD pool, Nonempty recovers 7.21 pp (four-model) and 3.58 pp (Qwen3/GLM) over Unfiltered without losing any successful Unfiltered answer, yet both pools leave about 9 pp of questions with a correct candidate unreturned; on Spider the Qwen3/GLM pool has slightly lower availability but higher Unfiltered accuracy (78.13% versus 77.02%) because fewer available answers go unused.

Splitting DeepSeek-V4-Pro generation's gain over Nonempty by availability (Section VIII-B): on BIRD, 8.16 of the 8.87 pp come from the 557 questions whose four-model pool holds no correct query, where generation answers 118 (21.2%), Qwen3-32B generation 59 (10.6%), and voting and both selectors none; on the 977 questions with a correct candidate, generation is $+0.72$ pp from Nonempty, not detected, and 1.51 pp below the DeepSeek-V4-Pro selector, detected. On Spider, generation newly answers 28 questions without a correct candidate ($+1.91$ pp), and losses on questions with one offset them, so the total is not detected.

TABLE S33
GENERATION MINUS VOTING ACROSS GENERATORS (PP).

	BIRD		Spider	
Recipient	Δ	95% CI	Δ	95% CI
Generation minus Unfiltered				
Qwen3	$+10.82^*$	$[8.25, 13.30]$	$+1.26$	$[-1.46, 4.07]$
gpt-oss	$+9.62^*$	$[7.40, 11.90]$	$+0.26$	$[-2.71, 3.09]$
Gemma	$+13.25^*$	$[10.36, 16.01]$	$+5.13^*$	$[1.80, 8.08]$
Generation minus Valid				
Qwen3	$+6.35^*$	$[4.26, 8.44]$	$+0.05$	$[-2.69, 2.69]$
gpt-oss	$+5.15^*$	$[2.47, 7.82]$	-0.94	$[-4.04, 2.10]$
Gemma	$+8.78^*$	$[6.63, 10.87]$	$+3.93$	$[0.85, 6.60]$
Generation minus Nonempty				
Qwen3	$+3.61^*$	$[1.84, 5.22]$	-0.43	$[-3.13, 2.12]$
gpt-oss	$+2.41$	$[-0.01, 4.57]$	-1.42	$[-4.39, 1.59]$
Gemma	$+6.04^*$	$[3.93, 8.20]$	$+3.44$	$[0.74, 5.91]$

Four candidates; database weights and intervals; *: passes joint Holm over 18 contrasts.

B. Generation Contributions across Candidate Pools

Table S39 expands Section VIII-B. Each stratum contributes to the total generation-minus-Nonempty effect using the same database weights; the entries are not conditional accuracies. The four-model pool is defined in Section VII-B; six candidates add Mistral-7B and Qwen2.5-72B. Qwen2.5 results average three generations per question; Qwen3 and the cross-family rows use the first response generated for each input, and DeepSeek uses one generation. The Qwen2.5 rows retain 1,533 or 1,532 BIRD questions with all three generation repeats; other rows use 1,534.

Every absent-candidate contribution is detected by both pointwise intervals; no present-candidate contribution is detectably positive. These strata use reference answers after execution. The four-model pool's 557 absent-candidate questions include two without a completed reference result, assigned zero correctness for every operation. Excluding them gives a DeepSeek generation gain of 8.88 pp, of which 8.16 pp comes from absent-candidate questions, preserving the interpretation.

TABLE S34
LLM SELECTOR WITH FULL INPUTS MINUS OTHER OPERATIONS, FIRST RESPONSE PER INPUT (PP).

Pool	Benchmark, selector minus	Effect	95% CI
Four-model	BIRD, generation	-2.26	$[-3.68, -0.53]$
	BIRD, selector (execution summary)	$+0.38$	$[-0.73, 1.45]$
	BIRD, Nonempty	$+1.35$	$[-0.30, 3.03]$
	Spider, generation	$+0.89$	$[-0.42, 2.30]$
	Spider, selector (execution summary)	-0.12	$[-1.22, 0.92]$
	Spider, Nonempty	$+0.46$	$[-1.93, 2.40]$
Qwen3/GLM	BIRD, generation	-0.04	$[-1.66, 1.50]$
	BIRD, selector (execution summary)	-0.51	$[-1.46, 0.59]$
	BIRD, Nonempty	-0.07	$[-1.25, 1.24]$
	Spider, generation	-0.03	$[-0.81, 0.80]$
	Spider, selector (execution summary)	-0.26	$[-1.15, 0.57]$
	Spider, Nonempty	-1.93^*	$[-3.68, -0.60]$

Database weights; six-contrast Holm family per pool; *: passes.

TABLE S35
GENERATION MINUS LLM SELECTION WITH FULL INPUTS BY GENERATION REPEAT (PP).

Benchmark	Pool	Repeat 1	Repeat 2	Mean
BIRD	Four-model	2.26	3.29	2.77
BIRD	Qwen3/GLM	0.04	0.89	0.46
Spider	Four-model	-0.89	0.25	-0.32
Spider	Qwen3/GLM	0.03	-0.29	-0.13

Means use unrounded effects.

TABLE S36
QWEN3/GLM POOL: TWO-REPEAT CONTRASTS AND THEIR CHANGE FROM THE FOUR-MODEL POOL (PP).

Benchmark	Contrast	Effect	95% CI
	Within the Qwen3/GLM pool		
BIRD	Generation − selector	0.46	$[-0.48, 1.48]$
BIRD	Generation − Nonempty	0.48	$[-0.64, 1.57]$
BIRD	Selector − Nonempty	0.02	$[-1.13, 1.32]$
Spider	Generation − selector	-0.13	$[-0.74, 0.50]$
Spider	Generation − Nonempty	-1.91^*	$[-3.35, -0.69]$
Spider	Selector − Nonempty	-1.78^*	$[-3.46, -0.47]$
	Change from the four-model pool		
BIRD	Generation − selector	-2.31^*	$[-3.67, -0.89]$
BIRD	Generation − Nonempty	-3.42^*	$[-5.56, -1.35]$
BIRD	Selector − Nonempty	-1.10	$[-3.02, 0.49]$
Spider	Generation − selector	$+0.19$	$[-1.29, 1.70]$
Spider	Generation − Nonempty	-1.81	$[-4.53, 1.05]$
Spider	Selector − Nonempty	-2.00	$[-3.68, -0.27]$

Selector: full inputs; two generation repeats per question; one Holm family per block; *: passes.

TABLE S37
CORRECT SQL CANDIDATES AVAILABLE BUT NOT RETURNED (%).

Candidate source	Available	Available, not returned	
		Unfiltered	Nonempty
	BIRD		
Four-model pool	63.36	16.25	9.03
Qwen3/GLM	66.62	12.80	9.22
	Spider		
Four-model pool	84.96	7.94	6.25
Qwen3/GLM	84.11	5.98	4.87

1,532 BIRD and 1,034 Spider questions; database weights.

TABLE S38
SQL QUALITY AND INCREMENTAL LLM COST PER OPERATION.

Operation	Acc. (%)	Input	Output	$ / 1k
BIRD, four-model pool				
Nonempty	54.32	0	0	0.000
Selector, execution summary	53.79	476	3	0.039
Selector, full inputs	55.44	1,077	1,435	0.488
Generation	58.22	1,066	1,067	0.384
BIRD, Qwen3/GLM pool				
Nonempty	57.40	0	0	0.000
Selector, execution summary	57.03	600	3	0.049
Selector, full inputs	57.41	1,196	1,357	0.476
Generation	57.88	1,185	935	0.357
Spider, four-model pool				
Nonempty	78.71	0	0	0.000
Selector, execution summary	76.97	379	3	0.031
Selector, full inputs	78.92	549	1,196	0.379
Generation	78.60	538	745	0.252
Spider, Qwen3/GLM pool				
Nonempty	79.24	0	0	0.000
Selector, execution summary	77.57	412	3	0.034
Selector, full inputs	77.46	582	1,094	0.353
Generation	77.33	571	717	0.246

Tokens per question; $ per 1,000 questions at $0.08/$0.28 per million input/output tokens.

TABLE S39
BIRD GENERATION GAINS BY CORRECT-CANDIDATE AVAILABILITY (PP).

Recipient	Candidate pool	Total	Correct candidate present	Correct candidate absent
DeepSeek-V4-Pro	Four-model	$+8.87$	$+0.72$	$+8.16$
Qwen3-32B	Four-model	$+3.61$	-0.01	$+3.62$
Qwen3-32B	Qwen3/GLM	-0.03	-0.75	$+0.72$
Qwen2.5-7B	Four-model	-4.50	-5.49	$+0.99$
Qwen2.5-32B	Four-model	$+1.04$	-0.92	$+1.96$
Qwen2.5-7B	Six candidates	-5.56	-6.75	$+1.19$
Qwen2.5-32B	Six candidates	$+0.09$	-0.93	$+1.02$
Gemma-4-26B	Four-model	$+6.04$	$+0.23$	$+5.82$
gpt-oss-120b	Four-model	$+2.41$	-2.43	$+4.85$

C. Backbone Scale and Cross-Family Debates

Table S40 gives every scale contrast behind Table X, and Table S41 the per-model values summarized in Section VIII-D with their intervals. All values use the lenient extractor, the execution vote, ties going to the first agent, and the same 198 BIRD questions; intervals are paired bootstraps over questions and, in a second set, over the 11 databases. Each per-model value is the per-task mean of Du-MAD and MAD-M^2, and later rounds means the last round minus the initial answers, with peers. The 7B point uses fp16 weights and the other Qwen2.5 points AWQ, and the Llama contrast also changes the model version. The configuration tested for equivalence, MAD-M^2 at 72B, has Δ_{peer} $+1.01$ pp with 90% CI $[-1.52, +3.79]$ against a ± 2 pp equivalence margin; its two repeats give $+3.03$ and -1.01 pp. The scale contrast is 72B minus 7B on the Du-MAD and MAD-M^2 mean, chosen with the per-configuration values in view, so its interval is descriptive; its 7B point is fp16 and is that of the protocol matrix. In the cross-family Holm family of nine models, every Δ_{peer} has $p_{\text{Holm}} = 1.0$ except MiniMax-M2.5 (0.909); the Qwen2.5 series, gpt-oss, and Llama-3-8B are tested in their own contrasts. For gpt-oss, MAD-M^2's perplexity filter reads the log-probabilities of the final channel. MiniMax-M2.5 runs Du-MAD only, so its row is a Du-MAD value. On gpt-oss-20B and 120B, GLM-4.7-Flash, Qwen3-32B, and Agentar-SQL-32B, BCCS's extractor accepts almost none of the answers, so its accuracy is near zero in every round and its round-to-round change is about zero by construction; it is omitted there. Nemotron-3.5-Bolt is excluded because 8.3% of its calls leak reasoning into the answer and one configuration truncates 14% of its responses.

Online check. Running BCCS online on Qwen2.5-7B with execution voting inside the protocol, which matches the replay only on average and ran on a different serving setup, gives the peer interaction effects in Table S42: -2.78 pp with ties to the first agent (database-cluster interval $[-7.44, +1.99]$), -1.01 pp when ties go to the last agent or at random, and -3.51 pp when the runs that stop on a single executing answer are excluded, none detected; the per-database differences are positive in four of the 11 databases. The online outcomes lie inside the lemma's interval for the replayed BCCS configurations in both conditions (Section S2-C).

Evaluator check. The scale and cross-family runs are scored by a second implementation of the same execution and

TABLE S40
SCALE CONTRASTS OF THE PEER INTERACTION EFFECT, LARGER MINUS SMALLER (PP).

Contrast	$\Delta\Delta_{\text{peer}}$ CI (questions)	CI (databases)	p (p_{Holm})
Qwen2.5, mean of Du-MAD and MAD-M^2 (five-contrast Holm family)			
72B − 7B (scale contrast)	$+0.51$ $[-2.40, +3.41]$	$[-1.62, +2.69]$	0.760 (1.000)
14B − 7B	-3.28 $[-7.32, +0.63]$	$[-5.65, -0.76]$	0.113 (0.446)
32B − 14B	$+3.03$ $[-0.63, +6.82]$	$[-0.88, +6.76]$	0.112 (0.446)
72B − 32B	$+0.76$ $[-2.15, +3.66]$	$[-2.64, +4.29]$	0.633 (1.000)
72B − 14B	$+3.79$ $[+0.63, +7.07]$	$[+1.80, +5.80]$	0.018 (0.092)
Other families (not in the Holm family)			
gpt-oss 120B − 20B	-1.77 $[-5.68, +2.02]$	$[-5.20, +1.55]$	0.386
Llama 3.3-70B − 3-8B	$+4.04$ $[+0.13, +7.95]$	$[+0.62, +7.14]$	0.049

TABLE S41
CROSS-FAMILY DEBATES ON BIRD: LATER ROUNDS WITH PEERS AND THE PEER INTERACTION EFFECT (PP).

Model	Later rounds 95% CI	Δ_{peer} 95% CI
Agentar-SQL-32B[h]	$+0.13$ $[-1.26, +1.64]$	$+0.63$ $[-1.52, +2.90]$
Gemma-4-26B-A4B[h]	-1.14 $[-3.91, +1.52]$	-1.52 $[-4.29, +1.14]$
GLM-4.7-Flash[h]	$+1.89$ $[-0.25, +4.04]$	$+2.15$ $[-1.01, +5.43]$
gpt-oss-120b	-2.15 $[-4.04, -0.38]$	-2.27 $[-5.05, +0.38]$
gpt-oss-20b	-4.55 $[-7.07, -2.15]$	-0.51 $[-3.41, +2.40]$
Llama-3-8B	-0.88 $[-3.03, +1.14]$	-3.91 $[-6.94, -0.88]$
Llama-3.3-70B[h]	-0.51 $[-2.27, +1.14]$	$+0.13$ $[-2.27, +2.40]$
MiniMax-M2.5[h, d]	$+6.31$ $[+3.03, +9.60]$	-4.29 $[-9.34, +0.76]$
Mistral-Small-3.2[h]	-1.01 $[-2.65, +0.63]$	-0.13 $[-3.03, +2.78]$
Qwen2.5-7B	$+0.13$ $[-2.02, +2.27]$	0.00 $[-2.90, +2.78]$
Qwen2.5-14B	-1.52 $[-3.66, +0.51]$	-3.28 $[-6.06, -0.63]$
Qwen2.5-32B	-0.88 $[-2.53, +0.76]$	-0.25 $[-2.78, +2.27]$
Qwen2.5-72B	-0.51 $[-1.64, +0.63]$	$+0.51$ $[-1.01, +2.15]$
Qwen3-32B[h]	-0.38 $[-2.40, +1.64]$	-1.26 $[-4.04, +1.39]$
Qwen3.6-35B-A3B[h]	-0.88 $[-2.65, +0.88]$	-1.14 $[-3.03, +0.76]$
Qwen3.8-27B[h]	-1.01 $[-2.78, +0.76]$	-1.01 $[-3.41, +1.39]$

Intervals over questions; [h]: cross-family Holm family; [d]: Du-MAD only.

result comparison. On the Llama-3-8B and Qwen2.5-7B runs (4,800 each, plus a 2,800-run mixed-model roster) the two implementations agree on all 12,400 runs; on a random sample of 250 runs per stronger model they agree on 245–250, every disagreement being an execution timeout in one of them. This is a sample check, not a full re-score.

D. Debate Budgets and Independent Drafts

These configurations and sensitivities extend Section VIII-E. The full-response baseline has three agents and two rounds. Comparisons use 198 BIRD questions and two debate repeats, with 20 independent initial drafts per question and backbone. Drafts use the protocol's initial prompt; Du-MAD decodes greedily and MAD-M^2 at temperature 1. A separate Du-MAD draft condition uses temperature 0.7. Each vote uses the first k drafts with first-agent ties; averaging over 20 draft orderings gives a sensitivity to ordering.

Tokens include prompt and completion tokens. Du-MAD's failures are rejected model requests; the combination of accumulated history and an 8,192-token context suggests a context-budget failure, but the returned error does not identify the cause. Conditional estimates average successful repeats and retain 155 paired questions for the four-round Du-MAD mean. Scoring failed runs as wrong changes the four-round-minus-base effect from -0.16 to -2.02 pp for Du-MAD and from $+1.05$ to -0.32 pp over both protocols. Independent drafts and the five-agent conditions have no failures. Unparsed or truncated outputs are scored as returned, with unparsed drafts treated as wrong.

Token matching chooses the smallest integer draft count whose mean tokens reach the debate's mean. For MAD-M^2, the Llama/Qwen counts are 7/8, 11/13, 13/15, and 14/17 for the four configurations in table order. The comparisons can overshoot by one draft. Du-MAD's base configuration uses 11 drafts; its larger budgets exceed 20 drafts and are compared against the cheaper 20-draft cap. Against greedy drafts, Du-MAD's call-matched differences are $+0.63$, $+1.28$,

TABLE S42
BCCS RUN ONLINE WITH EXECUTION VOTING, QWEN2.5-7B (PP).

Tie-breaking	Δ_{peer}	95% CI (questions)
First agent	-2.78	$[-6.82, +1.26]$
Last agent	-1.01	$[-5.05, +3.03]$
Random	-1.01	$[-4.80, +2.78]$
First agent, single-answer stops excluded	-3.51	$[-7.57, +0.54]$

TABLE S43
DEBATE CONFIGURATIONS, TOKEN COSTS, AND FAILURES.

Protocol	Backbone	Agents × rounds	Calls	Tokens per run	Failed runs
Du-MAD	Llama-3-8B	3×2	6	11,021	0
		3×3	9	22,881	0
		5×2	10	21,816	0
		3×4	12	36,878	33
Du-MAD	Qwen2.5-7B	3×2	6	11,760	0
		3×3	9	25,018	4
		5×2	10	23,368	0
		3×4	12	39,195	80
MAD-M^2	Llama-3-8B	3×2	6	6,604	0
		3×3	9	10,045	0
		5×2	10	12,653	0
		3×4	12	13,348	0
MAD-M^2	Qwen2.5-7B	3×2	6	7,753	0
		3×3	9	12,611	0
		5×2	10	15,082	0
		3×4	12	17,596	0

TABLE S44
MAD-M^2 DEBATE MINUS INDEPENDENT VOTING AT MATCHED BUDGETS (PP).

Budget	Debate configuration	Difference	CI, questions	CI, databases
Calls	3 agents, 2 rounds	-3.91	$[-7.20, -0.76]$	$[-8.08, +0.50]$
Calls	3 agents, 3 rounds	-4.17	$[-7.45, -0.88]$	$[-5.96, -2.44]$
Calls	5 agents, 2 rounds	-3.91	$[-7.07, -0.88]$	$[-6.31, -1.62]$
Calls	3 agents, 4 rounds	-5.56	$[-8.59, -2.65]$	$[-7.49, -3.39]$
Tokens	3 agents, 2 rounds	-6.19	$[-9.60, -2.90]$	$[-9.88, -2.12]$
Tokens	3 agents, 3 rounds	-3.16	$[-6.31, +0.00]$	$[-4.60, -1.78]$
Tokens	5 agents, 2 rounds	-4.42	$[-7.58, -1.39]$	$[-7.46, -1.49]$
Tokens	3 agents, 4 rounds	-6.06	$[-9.09, -3.03]$	$[-8.21, -3.89]$

$+0.38$, and $+0.32$ pp, none detected. Against temperature-0.7 drafts, they are -4.92, -4.59, -5.18, and -6.13 pp, all detected; this comparison changes decoding.

TABLE S45
DIVERSITY AND ACCURACY GAINS FROM INDEPENDENT DRAFTS.

Protocol	Backbone	Temperature	Distinct results	Multiple results (%)	Vote@20 − vote@1 (pp)
Du-MAD	Llama-3-8B	0	0.82	9.6	0.00
Du-MAD	Qwen2.5-7B	0	0.90	10.6	+2.53
Du-MAD	Llama-3-8B	0.7	3.19	71.7	+10.61
Du-MAD	Qwen2.5-7B	0.7	3.13	72.7	+7.58
MAD-M^2	Llama-3-8B	1	3.21	75.3	+15.15
MAD-M^2	Qwen2.5-7B	1	3.39	74.7	+15.15

Distinct results counts executable result sets among 20 drafts, averaged across questions; a question with no executable query contributes zero. Textually different greedy replies can therefore offer the same result. The temperature-0.7 and temperature-1 vote gains are detected, while the two greedy gains are not.

The six-draft comparison in the first paragraph of Section VIII-E uses independent initial responses from two repeats of the sampled debate collection. Under a common lenient extractor and execution vote, its four-configuration vote-minus-debate effect is $+1.33$ pp, compared with $+10.10$ pp against the protocols' final answers. This fixed-response comparison and the separately executed budget configurations use different draft collections and are not pooled. The underlying stage-removal comparisons in Table VII compare votes over each run's own initial answers and therefore use fewer calls than the complete debate.